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The impact of analyst coverage on partial acquisitions: Evidence from M&A premium and firm performance in China

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ABSTRACT

In this paper, we examine how analyst coverage affects mergers and acquisitions in China. By using partial acquisitions in our study, we are able to directly examine the post-acquisition performance of the targets. Our results are robust and consistent with our hypotheses. Analyst coverage can effectively reduce information asymmetry. As a result, targets with high analyst coverage experience more immediate and accurate price correction in the short run, as indicated in the acquisition premium. In the long run, targets with high analyst coverage outperform targets with low analyst coverage up to two years post acquisition, based on EPS. Our results indicate analyst coverage can effectively (1) speed up price discovery by bringing targets' stock prices closer to their equilibria, (2) promote market efficiency by reducing information asymmetry, (3) help acquirers make better asset allocation and investment decisions in the acquisition market by reducing information asymmetry, and (4) support targets' long-term performance by providing them with better access to external resources.

1. Introduction

Analysts play the role of information intermediary in the financial markets. They conduct research on their covered firms and disseminate their findings to the public. In the process, they lower the information asymmetry between the firm and investors and to monitor the performance of the firms. Few studies, however, examine the role of analyst coverage in mergers and acquisitions. Specifically, the impact of targets' analyst coverage on their acquisition premium in partial acquisitions has not yet been examined.

The objective of this paper is to examine how analyst coverage affects acquisition premium and post-acquisition performance of targets in partial acquisitions using a sample of Chinese firms. To make sure that we can directly examine the post-acquisition performance of the targets, rather than of the merged firms, we use partial acquisitions in our study (since these targets will maintain their listings post partial acquisition). In addition, our study is interesting because China is one of the largest emerging markets in the world and is known to have weaker investor protection than developed countries and common law countries (Allen, Qian, & Qian, 2005; La Porta et al., 1998). Furthermore, Chinese firms tend to have more information asymmetry (Piotroski, Wong, & Zhang, 2015). Therefore, we argue that analysts play an important role in the economy by reducing information asymmetry, bringing stock prices close to their equilibria, and promoting market efficiency. Our partial acquisition data in China provide us an excellent opportunity to examine how analyst coverage affects acquisition premium and post-acquisition performance of the target firms. We hypothesize that analyst coverage reduces information asymmetry. Since firms in China tend to have higher growth, stocks are more likely to be overvalued as a result of

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Table 1
Variable selection.
Panel A presents our sample selection process. Panel B presents the distribution of the sample by year from 2004 to 2016. Column 2 reports the number of partial acquisitions for each year. Column 3 reports the percentage of acquisition in each year with respect to the total sample from 2004 to 2016. Panel C presents the distribution of acquisition percentage in our partial acquisition sample.

Panel A: Sample Selection	
Initial sample: All Chinese A-share target companies listed on Shanghai Stock Exchange and Shenzhen Stock Exchange are obtained from the CSMAR ″M&A″ database from its data inception date in 2004–2016. Open market transactions are excluded from the sample.	6044
Observations are eliminated when any of the following is true:	
The transfer price per share is unknown or 0	3729
It is listed as a "Special Treatment" (ST) stock. Under regulations of Shenzhen Stock Exchange and Shanghai Stock Exchange, when the listed company experiences financial issues or other abnormal conditions, the Stock Exchanges will mark the stock as a ST stock until the normal conditions return.	9
The observation is a financial company	144
Any of the control variables is missing	902
Failed acquisition	53
Final sample	1207

Panel B: Sample distribution by year

Year	Total Sample	Ratio
2004	89	7.37%
2005	105	8.70%
2006	113	9.36%
2007	53	4.40%
2008	40	3.31%
2009	57	4.72%
2010	34	2.82%
2011	25	2.07%
2012	54	4.47%
2013	44	3.65%
2014	46	3.81%
2015	373	30.90%
2016	174	14.42%
Total	1207	100%

Panel C: Description of Partial Acquisition

	Mean	Min	P25	Median	P75	Max	Std. dev.
Acquisition Percentage	0.1041	0.0001	0.0056	0.0600	0.1587	0.5963	0.1242

Table 2
Descriptive statistics.
This table presents the descriptive statistics for the variables used in the study. Please refer to Appendix A for the detailed definitions and calculations. All variables are winsorized at the 1 and 99 percentiles.

Variables	Mean	Min	P25	Median	P75	Max	Std. dev.
Premium1	−0.2248	−0.9096	−0.4789	−0.1803	−0.0099	0.7500	0.3435
Premium2	2.4807	−0.9640	−0.0203	0.9725	3.0207	44.2993	5.5946
AnaDummy	0.5667	0	0	1	1	1	0.4957
AnaTimes	9.6694	0	0	1	11	78	17.0719
AnaNum	4.8376	0	0	1	6	37	7.8518
EPS	0.2251	−1.9618	0.0369	0.1538	0.3945	2.1760	0.5328
Transparency1	−0.2079	−1.0902	−0.2565	−0.1598	−0.0956	−0.0265	0.1746
Transparency2	−0.0671	−0.4597	−0.0853	−0.0445	−0.0190	−0.0004	0.0758
Adjust	−0.0028	−0.0729	−0.0103	−0.0009	0.0060	0.0584	0.0181
CashPay	0.9395	0	1	1	1	1	0.2385
Relevance	0.1152	0	0	0	0	1	0.3193
MTB_Asset	2.4719	0.9309	1.3681	1.8594	2.8816	11.6624	1.8222
MTB_Equity	4.1463	0.8137	1.8609	2.9038	4.5457	36.4870	4.7992
Growth	0.1954	−0.6478	−0.0552	0.1028	0.2953	3.2794	0.5524
Leverage	0.4839	0.0560	0.3131	0.4967	0.6559	0.9379	0.2180
LnAsset	21.6450	19.2360	20.8090	21.5245	22.3015	24.9742	1.1725
Share	0.3361	0.0880	0.2299	0.3037	0.4278	0.7064	0.1408
ROA	0.0206	−0.3255	0.0063	0.0258	0.0525	0.1944	0.0750
OverConfidence	0.4186	0.1741	0.3179	0.3927	0.4906	0.9032	0.1401
Cashflow	0.0429	−0.2094	0.0025	0.0397	0.0881	0.2688	0.0768
SDAR	0.0255	0.0123	0.0202	0.0251	0.0302	0.0434	0.0069
CAR	0.0103	−0.4038	−0.0978	−0.0019	0.0947	0.5305	0.1700
Proportion	0.3621	0	0	0	1	1	0.4808
Precision	0.7387	0	0	1	1	1	0.4397

investors' over-optimism. If analyst coverage can successfully reduce information asymmetry, targets with high analyst coverage should earn lower acquisition premium when price correction takes place. In addition, the reduction of information asymmetry will help acquirers make better investment decisions and targets receive the resources they need. Consequently, targets with higher analyst coverage should outperform those with lower analyst coverage post acquisition. Our findings are consistent with our hypotheses, while our results remain robust after a battery of robustness checks.

We provide two contributions in this study. First, while majority of the analyst coverage studies have examined the impact of analyst coverage on the financial markets (Schutte & Unlu, 2009; Xu, Chan, Jiang, & Yi, 2013), we are able to extend the growing literature on how analyst coverage influences market efficiency and corporate investment decisions to the M&As in China. Our first contribution has multiple layers: (1) In M&As, acquirers are much more sophisticated and experienced than the average investors in the market. If analyst coverage is able to significantly reduce information asymmetry for acquirers during M&A negotiations, it can have even more incredible impact and influence for the market investors; (2) While many papers have examined the M&A premium in the United States, we are the first paper that examines the acquisition premium in China, the largest emerging market in the world. Since China is well known for its weaker investor protection and high information asymmetry, analyst coverage plays an extremely important role on improving market efficiency. Therefore, our data allow us to fill out the gap between the M&A and the analyst coverage literature and extend our understanding to the emerging markets. Second, unlike majority of the M&A studies, we use partial acquisition data in our study. By doing so, we are able to directly examine the post-acquisition performance of the target firms, rather than of the merged firms. As an additional benefit, the partial acquisition sample allows us to examine earnings per share (EPS) performance that is otherwise extremely challenging to measure with the merged firm sample in most M&A studies. Therefore, rather than focusing on the post-acquisition stock and operating performance as in most M&A studies (Jory, Ngo, & Wang, 2016; Loughran & Vijh, 1997), we take this opportunity to examine the post-acquisition EPS of the targets.

Our results indicate that analyst coverage can: (1) speed up price discovery and bring targets' stock prices closer to their fair market values, (2) promote market efficiency by reducing information asymmetry in the emerging markets, (3) improve acquirers' asset allocation when making investment decisions in the acquisition market,[1] and (4) support targets' post acquisition performance by reducing information asymmetry and hence provide targets with external financial and non-financial resources.

2. Literature review

In this section, we will provide two strands of literature review: analyst coverage and mergers and acquisitions.

2.1. Analyst coverage

It has been well documented that analyst coverage can reduce information asymmetry. Information asymmetry creates noise and mispricing when market participants look at the wrong information or draw the wrong inferences from the available information. Schutte and Unlu (2009) use residual volatility to measure noise in the stock price and conclude that analyst coverage reduces noise. Sadka and Scherbina (2007) find higher analyst disagreement to be positively correlated with mispricing, while less liquid stocks tend to be more mispriced. Feng (2007) finds that analysts' prediction accuracy is significantly better than the random walk model. Feng, Hu, and Johasson (2016) examine the effect of ultimate ownership structure and analyst coverage on stock return synchronicity in China and find that stock return synchronicity to analyst coverage tends to increase when the firm has greater separation of control and ownership rights. Xu et al. (2013) find that star analysts are more successful at overcoming the information barrier in emerging markets due to their superior firm-specific human capital. As a result, star analysts provide more accurate earnings forecasts than non-star analysts.

A few studies have examined the impact of analyst coverage on firm performance. Das, Guo, and Zhang (2006) use IPO firms to proxy for information asymmetry and find that higher analyst coverage is associated with better future performance for IPO firms. Lang, Lins, and Miller (2004) find higher analyst coverage leads to higher valuation. Bowen, Chen, and Cheng (2008) find higher analyst coverage reduces cost of capital, while Roulstone (2003) finds analyst coverage to improve market liquidity. Barber, Lehavy, McNichols, and Trueman (2006) show that changes in analyst' recommendations can predict stock returns, while Irvine (2003) finds positive price impact on firms with analyst initiations. Jung (2017) finds a U-shaped relationship between analyst coverage and innovation. Initially, innovation of the firm declines when analyst coverage increases moderately. However, as analyst coverage continues to increase, innovation increases thereafter. Dhiensiri and Sayrak (2004) examine stocks that have never had analyst coverage in the past and find firms with first time initiations by analyst coverage earn an average of about 2% abnormal returns at announcement.

2.2. Mergers &acquisitions

The existing short-term M&A studies often use market reactions to indirectly measure acquisition premium. It has been well documented that acquirers of domestic M&As in general earn either negative or insignificant returns at announcement (Andrade, Mitchell, and Staffors, 2001; and Moeller, Schlingmann, and Stulz, 2004). Officer, Poulsen, and Stegemoller (2009) find more positive market reactions when acquirers use stock to acquire targets with more information symmetry problems. Travlos (1987), Houston and Ryngaert (1997) and Fuller, Netter, and Stegemoller (2002) find that acquirers on average earn significantly negative (statistically

[1] Stock price stability affects how managers make decisions from stock issuance, stock repurchase, dividend payout policy, investment, to acquisitions and divestitures (Schutte & Unlu, 2009).

insignificant) returns when acquiring public firms with stock (cash) payment.

Target firms in the United States have been well documented to earn significantly positive premium (Hayward & Hambrick, 1997; Walking & Edmister, 1985; and Varaiya & Ferris, 1987). Kohers and Kohers (2001) report that targets in the United States earn 33.4%, 38.4%, and 44.0% premia for the 1-day, 1-week, and 4-week event windows, while technology industry earns the highest acquisition premium. Niden (1988), Franks and Harris (1989), and Asquith, Bruner, and Mullins (1990) find that targets tend to earn higher premium when acquirers pay cash for targets' stocks. Bradley, Desai, and Kim (1988) find a positive relationship between target shareholders' returns and the percentage of target shares acquired. Niden (1988) finds an inverse relationship between the target's pre-acquisition market-to-book equity ratio and acquisition premium. Franks and Harris (1989) find a negative relation between acquisition premium and the relative size of the target. Comment and Schwert (1995) find tender offers tend to earn higher acquisition premiums, for target management often initially resist tender offers.

Long-term M&A studies post successful mergers and acquisitions are often used to determine whether acquirers make good M&A decisions. The long-term stock underperformance indicates that acquirers often make suboptimal decisions because of managerial hubris and overconfidence (Roll, 1986; Loderer & Martin, 1992; Loughran & Vijh, 1997; and Gregory, 1997). Lin et at. (2011) examine the long-term performance of stock financed acquirers and find that acquirers with overvalued stocks at the time of acquisition tend to underperform in the long run. The underperformance is the result of price correction for their originally overvalued stocks. Hardford (1999) finds that cash-rich firms are more likely to attempt acquisitions, while merged firms with cash-rich acquirers tend to underperform in the long run both in stock return and in operating performance. Jory et al. (2016) find the bidders who acquire targets with credit rating tend to have better operating performance post acquisition than those who acquire targets without credit ratings. Their results are consistent with the hypothesis that if there is credible information (e.g., credit ratings) available to reduce information asymmetry, acquirers are more likely to pay fair market value for the targets.

A few studies have examined partial acquisitions, where the acquirers do not acquire 100% of the targets. Allen and Phillips (2000) and Fee, Hadlock, and Thomas (2006) find that partial acquisitions can mitigate incomplete contracts and facilitate cooperation between two firms. Fee et al. (2006), Liao (2013), and Hertzel and Smith (1993) find financially constrained targets are more likely to be associated with partial acquisitions. Bostan and Spatareanu (2018) examine the partial acquisitions in the Unites States and find that targets tend to experience increased cash flows and innovation post acquisition. On the other hand, targets without cash flow increase do not become more innovative. Ouimet (2013) finds acquirers tend to prefer partial acquisitions when: (1) maintaining target managerial incentives is important, (2) the target is financially constrained or can benefit from the M&A certifications,[2] (3) target's valuation is difficult, (4) integrating internal capital markets is costly, and (5) and consolidating earnings will deduce the EPS of the merged firms. Ouimet (2013) also argue that majority acquisitions can reduce target managers' bargaining power and their incentive to invest in relationship-specific assets, while partial acquisitions can facilitate the flow of information between two independent firms.

3. Hypotheses

We propose our hypotheses in an environment of high information asymmetry with a lot of noise and significant mispricing (Diamond & Verrechia, 1991). While there are many market participants, investors often look at the wrong information or draw the wrong inferences from the information for information asymmetry creates noise (Schutte & Unlu, 2009). Noise inflicts cost on market participants when it pushes stock price away from its equilibrium (De Long, Shleifer, Summers, & Waldmann, 1990), while financial analysts can help investors to distinguish between relevant information from noise. In the case with acquisitions, analyst's noise-reducing ability helps corporate managers make better asset allocations and investment decisions, while promoting market efficiency.

In our first hypothesis, we contend that *analyst coverage will reduce noise created by information asymmetry and bring the stock price of the target closer to its fair market value in a timely matter*. In the Chinese market setting, high information asymmetry (combined with very high growth and significant mispricing) predicts that if analyst coverage can speed up price discovery, targets are likely to be acquired at prices below their market value prior to the first acquisition announcement because their stocks are overvalued at the first place. Since acquirer's stocks are also likely to be overvalued, we predict that acquirers are more likely to use cash payment when acquiring targets at the offering price below the previous stock price. Note that in the sample with partial acquisition, targets may choose to sell a portion of the firm at seemingly lower price when (1) partial acquisition can promote cooperation between two companies, (2) targets are facing financial constraints, (3) such acquisition can provide targets with access to acquirers' non-financial resources, or (4) acquirers can help provide certification effect to the targets to signal higher quality (Hertzel & Smith, 1993; Liao, 2013; Ouimet, 2013; and Fee et al., 2006).

In our second hypothesis, we hypothesize that *targets with higher analyst coverage should outperform those with lower analyst coverage post acquisition, if the lower acquisition premium from the first hypothesis is the result of price correction rather than price discount for poor performers*. If our second hypothesis is valid, the result will imply that the reduced information asymmetry between acquirers and targets allows acquirers to identify suitable targets and their fair market values more efficiently. In addition, reduced information asymmetry also allows target firms to obtain the external resources they need for long-term growth, while such resources are otherwise unobtainable in the high information asymmetry environment.

If both of our hypotheses are valid, together, analyst coverage provides several benefits to the market for it: (1) speeds up price discovery by bringing targets' stocks closer to their equilibria, (2) improves market efficiency by reducing information asymmetry, (3) helps acquirers to make better asset allocation and investment decisions, and (4) provides targets with better access to both financial and

[2] Certification is proxied by low analyst coverage and the absence of a preexisting block holder.

Table 3
Firm Characteristics based on Analyst Coverage.
In this table, all targets are separated into high analyst coverage and low analyst coverage subsamples. The difference and *P*-value of the difference are presented in columns 4 and 5. All variables are winsorized at the 1 and 99 percentiles. ***, **, and * denote significance at the 1%, 5% and 10% levels, respectively.

Variables	High Analyst Coverage (1)	Low Analyst Coverage (2)	Difference (1)–(2)	*P*-value
	AnaDummy = *1*	*AnaDummy* = *0*		
MTB_Asset	2.7109	2.1594	0.5515***	<0.001
MTB_Equity	4.0426	4.2819	- 0.2393	0.3908
Growth	0.2517	0.1217	0.1300***	<0.001
Leverage	0.4427	0.5378	- 0.0951***	<0.001
LnAsset	22.0785	21.0780	1.0005***	<0.001
ROA	0.0455	−0.0119	0.0574***	<0.001
Cashflow	0.0548	0.0273	0.0275***	<0.001
	AnaTimes > Mean	*AnaTimes* ≤ Mean	Difference (1)–(2)	*P*-value
MTB_Asset	2.9184	2.2954	0.6230***	<0.001
MTB_Equity	4.3148	4.0797	0.2351	0.4433
Growth	0.2838	0.1604	0.1234***	<0.001
Leverage	0.4302	0.5052	−0.0750***	<0.001
LnAsset	22.4080	21.3433	1.0647***	<0.001
ROA	0.0617	0.0044	0.0573***	<0.001
Cashflow	0.0595	0.0363	0.0232***	<0.001
	AnaNum > Mean	AnaNum ≤ Mean	Difference (1)–(2)	*P*-value
MTB_Asset	2.8834	2.2865	0.5969***	<0.001
MTB_Equity	4.2385	4.1047	0.1338	*0.6544*
Growth	0.2738	0.1600	0.1138***	<0.001
Leverage	0.4350	0.5060	−0 .0710***	<0.001
LnAsset	22.3988	21.3052	1.0936***	<0.001
ROA	0.0601	0.0028	0.0573***	<0.001
Cashflow	0.0573	0.0364	0.0209***	<0.001

Table 4
Univariate test of analysts coverage on M&A premium.
This table presents univariate test results of analyst coverage on M&A premium. The difference between the two subsamples and *P*-value of the difference are presented in columns 4 and 5. All variables are winsorized at the 1 and 99 percentiles. ***, **, and * denote significance at the 1%, 5% and 10% levels, respectively.

	High Analyst Coverage (1)	Low Analyst Coverage (2)	Difference (1)–(2)	*P*-value
	AnaDummy = *1*	*AnaDummy* = *0*		
Premium1	−0.1652	−0.1287	−0.0365**	0.0220
	AnaTimes > Mean	*AnaTimes* ≤ Mean		
Premium1	−0.1432	−0.1120	−0.0312**	0.0419
	AnaNum > Mean	*AnaNum* ≤ Mean		
Premium1	−0.1409	−0.1072	−0.0337**	0.0232

non-financial resources externally.

4. Data, methods, and results

We obtain domestic mergers and acquisitions data between January 1, 2004 and December 31 of 2016 from the China Stock Market and Accounting Research database (CSMAR). Our Chinese acquisition data serve as an excellent candidate for this study since emerging markets, such as China, have higher information asymmetry problems than developed markets. Note that while some may argue that the higher information asymmetry is mainly the result of weaker government regulation and investor protection (Morck, Yeung, & Yu, 2000), others have argued that sometimes firms choose to remain opaque to preserve their competitive edge (Ederington & Yawitz, 1985; Covitz & Harrison, 2003; and Krisgen, 2006; 2009). Consequently, analyst coverage should have more significant impact in China than mature markets. Furthermore, our unique sample of partial acquisitions allows us to directly examine targets' post-acquisition performance, while the typical M&A samples will cause targets to be delisted post acquisition. Since we use the actual offer price instead of market reactions to directly examine how analyst coverage affects acquisition premium, we do not include open market transactions in our sample.[3]

In Panel A of Table 1, we start with all Chinese targets with A-share stocks listed on Shanghai and Shenzhen Stock Exchanges and end up with an initial sample of 6044 target events with partial acquisition. We focus on partial acquisitions in our study, so that when the targets maintain their listing after acquisitions, we can directly measure their post-acquisition performance. After deleting observations

[3] More details on the different acquisition methods are discussed in Table 8.

Table 5
Multivariate analysis of analysts coverage on M&A premium.
This table presents the multivariate analysis of analyst coverage on M&A premium. All control variables are provided in the left column. Three analyst coverage measures (*AnaDummy*, *AnaTimes*, *and AnaNum*) are used, while results are provided in columns (1), (2), and (3), respectively. Only successful acquisitions are used. All variables are winsorized at the 1 and 99 percentiles. ***, **, and * denote significance at the 1%, 5% and 10% levels, respectively.

Control Variables	*Premium2*		
	(1)	(2)	(3)
AnaDummy	−0.0382**		
	(-2.1204)		
AnaTimes		−0.0012**	
		(-2.1075)	
AnaNum			−0.0029**
			(-2.1975)
CashPay	−0.0187	−0.0219	−0.0218
	(-0.6156)	(-0.5362)	(-0.5345)
Relevance	−0.0861***	−0.0886***	−0.0889***
	(-3.4606)	(-2.9366)	(-2.9473)
MTB_Asset	0.0020	0.0022	0.0024
	(0.4845)	(0.2494)	(0.2705)
MTB_Equity	−0.0038*	−0.0034	−0.0033
	(-1.7612)	(-1.3637)	(-1.3319)
Growth	−0.0271*	−0.0281*	−0.0279*
	(-1.7474)	(-1.9041)	(-1.8817)
Leverage	−0.0995	−0.0961*	−0.0994*
	(-1.6048)	(-1.7165)	(-1.7755)
LnAsset	0.0348***	0.0379***	0.0394***
	(3.7830)	(3.3595)	(3.4245)
Share	−0.1618**	−0.1773***	−0.1772***
	(-2.1310)	(-3.0423)	(-3.0446)
ROA	0.1126	0.1202	0.1265
	(0.6386)	(0.9935)	(1.0437)
OverConfidence	0.0529	0.0477	0.0441
	(0.8572)	(0.8015)	(0.7410)
Cashflow	−0.1636*	−0.1700	−0.1680
	(-1.7794)	(-1.4430)	(-1.4293)
SDAR	−5.7834***	−5.7357***	−5.7190***
	(-3.2936)	(-4.0321)	(-4.0182)
CAR	0.1901***	0.1933***	0.1926***
	(5.8939)	(3.6655)	(3.6561)
Intercept	−0.7625***	−0.8452***	−0.8742***
	(-3.2444)	(-3.1017)	(-3.1492)
N	1207	1207	1207
adj.R^2	0.356	0.347	0.347

with unknown offer price, listed as “special treatment” stocks, in the financial industry, with missing control variables, and listed as a failed acquisition, our final sample consists of 1207 target events. In Panel B of Table 1, we notice a sharp increase of acquisitions in 2015 and 2016, while 30.90% of the acquisitions in our sample are from 2015. In Panel C of Table 1, we find that targets in our sample sell an average of 10.41% of their firms, while the median target sells only 6% of the target during partial acquisitions.

In Table 2, we provide the descriptive statistics of the variables in our study, while the detailed definitions of these variables are presented in Appendix A. We notice several interesting findings in Table 2. Please note that while the majority of the existing acquisition premium studies use market reactions to indirectly measure acquisition premium, we use the actual offer price in our paper to provide more direct and reliable results. *Premium1* shows that more than 75% of the target shares are acquired at offer prices below their stock prices from four weeks before the first acquisition announcement. Note that we follow the literature (Schwert, 1996; Reuter, Tong, & Wu, 2012; and Jory et al., 2016) to calculate the acquisition premium, *Premium1,* as (offer price per share – closing price four weeks before the announcement date)/the closing price four weeks before the announcement date. In this case, the firm's stock price four weeks before the first announcement is used in *Premium1* as the base price to control for potential price movement from information leakage. For robustness checks, we also include an alternative acquisition premium measure, *Premium2*, as (offer price per share - net assets per share)/net assets per share, where net assets per share is used as the base price (Brau, Sutton, & Hatch, 2010; Laamanen, 2007; Chen & Li, 2016; and Jory et al., 2016). The mean of *Premium2* is 2.4807, indicating that, on average, our targets are acquired at the offer price that is 248.07% higher than its net asset value. Combined with the mean of −22.48% for *Premium1*, our results suggest that while *Premium1* shows targets in China on average are acquired at offering prices below their previous stock prices from four weeks earlier, the same offering price is, on average, 248.07% higher than its net asset value.

It is also very important to note that the market-to-book value of equity (*MTB_Equity*) has a mean of 4.1463, indicating that, on average, targets' stocks are traded at 414.63% of their book values, while the market-to-book value of asset (*MTB_Asset*) is also high with an average of 2.4719. Together with the 19.54% average of *Growth* in revenue, our results are consistent with our first hypothesis that

Table 6
Analyst coverage and M&A premiums: 2SLS regression.
This table presents 2-stage least square regression results of how analyst coverage affects M&A premium. Only successful acquisitions are used. In the first stage of the two-stage regression, $E\_Ana_{i,t}$ is used as the instrumental variable to estimate $AnaNum_{i,t}$, the number of analysts who follow target i at a given year. The estimated $AnaNum_{i,t}$ ($Pred\_ AnaNum_{i,t}$) from stage one is then used as the dependent variable in stage two of the two-stage least square regression analysis. *T*-statistics are provided in parentheses. ***, **, and * denote significance at the 1%, 5% and 10% levels, respectively.

Control Variables	Stage 1	Stage 2
	AnaNum	Premium1
Pred_AnaNum		−0.0027**
		(-2.0685)
CashPay	−0.0147	−0.0204
	(-0.7136)	(-0.9253)
Relevance	−0.0830	−0.0895*
	(-1.6425)	(-1.8346)
MTB_Asset	0.0019	0.0022
	(0.2674)	(0.3312)
MTB_Equity	−0.0030	−0.0032
	(-1.3442)	(-1.4777)
Growth	−0.0284*	−0.0267*
	(-1.8498)	(-1.7802)
Leverage	−0.1074	−0.1035
	(-1.1296)	(-1.1195)
LnAsset	0.0375**	0.0374***
	(2.6580)	(2.6528)
Share	−0.1414*	−0.1570**
	(-2.1121)	(-2.4769)
ROA	0.0995	0.1175
	(1.1901)	(1.3872)
OverConfidence	0.0575	0.0443
	(0.7541)	(0.5731)
Cashflow	−0.1646	−0.1705*
	(-1.6377)	(-1.7877)
SDAR	−5.4672***	−5.6936***
	(-3.4391)	(-3.8217)
CAR	0.1460**	0.1851***
	(2.8467)	(3.3153)
E_Ana	−0.0028**	
	(-2.2105)	
Intercept	−0.8880**	−0.5718*
	(-2.7304)	(-1.7896)
N	1207	1207
adj.R^2	0.378	0.366

Chinese firms tend to have very high growth. Therefore, Chinese stocks are more likely to be overvalued on average. While *Premium1* shows that the offer price, on average, is lower than the price from four weeks before the first announcement, it is still much higher than the firm's net assets, as indicated by *Premium2*. Therefore, our results in Table 2 are consistent with the logic in our first hypothesis, i.e., analyst coverage can reduce information asymmetry and promote market efficiency by bringing stock prices closer to their equilibria through acquisitions.

We use three analyst coverage measures (*AnaDummy*, *AnaTimes*, and *AnaNum)* in this paper for more reliable and robust results, since each variable may suffer from different weaknesses: (1) *AnaDummy*, which looks at whether the target has any analyst coverage at all during the 360-day period prior to the first acquisition announcement (McNichols & O'Brien, 1997; James & Karceski, 2006; Chan & Hameed, 2006; and Yu, 2008), (2) *AnaTimes*, which is the total number of earnings forecasts issued by all analysts for the target during the 360-day period prior to the first acquisition announcement, and (3) *AnaNum*, which is the total number of analysts who issue any earnings forecast for the target company during the 360-day period prior to the first announcement. Our results show that 56.67% of our sample firms have analyst following, while on average each firm receives 9.67 forecasts per year and is followed by 4.84 analysts, respectively. Two transparency measures are used here. *Transparency1* is calculated with three years of information before the event date, while *Transparency2* is calculated with only one year of information before the event date. The means of *Transparency1* and *Transparency2* are −0.2079 and −0.0671, respectively. The negative measures suggest that Chinese firms tend to have low transparency and high information asymmetry, while the difference between the two measures indicates that targets tend to increase their transparency before acquisition.

CashPay has a mean of 0.9395, indicating that 93.95% of our targets are paid with cash, rather than stock, which is consistent with the implication of our hypothesis. Potentially, acquirers have to pay cash to entice targets to sell at seemingly lower prices from four weeks before the announcement. Also note that if the target receives stock payment from the acquirer, the target may suffer when the acquirer's stock price falls in the future. In addition, in the high growth emerging markets, where most assets are overvalued, targets

Table 7
Robustness check with alternative premium measure.
This table presents the multivariate analysis of analyst coverage on an alternative premium, Premium2. All control variables are provided in the left column. Three analyst coverage measures (*AnaDummy*, *AnaTimes*, *and AnaNum*) are used, while results are provided in columns (1), (2), and (3), respectively. Only successful acquisitions are used. All variables are winsorized at the 1 and 99 percentiles. *T*-statistics are provided in parentheses. ***, **, and * denote significance at the 1%, 5% and 10% levels, respectively.

Control Variables	*Premium2*		
	(1)	(2)	(3)
AnaDummy	−0.5116**		
	(-2.3733)		
AnaTimes		−0.0172**	
		(-2.8544)	
AnaNum			−0.0411**
			(-2.5964)
CashPay	0.3594	0.3569	0.3541
	(0.9593)	(1.0007)	(0.7911)
Relevance	−0.3282	−0.3869	−0.3881
	(-0.6486)	(-0.9378)	(-1.3655)
MTB_Asset	0.9534*	0.5681	0.5692*
	(1.8982)	(1.5856)	(1.7963)
MTB_Equity	0.6149	0.5841***	0.5843***
	(1.6522)	(3.5900)	(3.0165)
Growth	−0.4166	−0.1913	−0.1918
	(-1.1235)	(-0.7184)	(-0.4373)
Leverage	−6.0118**	−2.9395**	−2.9149***
	(2.7420)	(2.2900)	(3.0173)
LnAsset	−0.2462	0.0053	0.0128
	(-1.1758)	(0.0440)	(0.0897)
Share	−0.7517	−0.8313	−0.8202**
	(-0.6690)	(-0.7650)	(-2.5690)
ROA	4.0891	4.0809	4.1090
	(1.1337)	(1.2359)	(1.5999)
OverConfidence	−0.4626	−0.6527	−0.6970
	(-0.4316)	(-0.6552)	(-0.7853)
Cashflow	−0.8550	−0.0732	−0.0705
	(-0.3224)	(-0.0278)	(-0.0562)
SDAR	−18.5710	−1.4439	−1.5614
	(-0.8464)	(-0.0867)	(-0.0864)
CAR	2.6333***	2.7966***	2.7829***
	(3.7359)	(5.0576)	(5.1978)
Intercept	0.5630	−3.2976	−3.3984
	(0.1403)	(-1.2204)	(-0.8875)
N	1207	1207	1207
adj.R^2	0.380	0.449	0.448

should be less inclined to receive acquirers' overvalued stocks as payment unless other forms of benefits are involved.[4] The highest return on assets (*ROA*) is 19.44% while an average target only earns 2.06% *ROA* per year. Potentially, the low ROA is the result of businesses reinvesting earnings to take advantage of the high growth opportunities.

In Table 3, we examine firm characteristics by separating targets into high vs. low analyst coverage subsamples. We find that targets with high analyst coverage tend to be more overvalued based on *MTB_Asset*; and have higher growth, lower financial leverage, larger firm size, better performance (as measured by *ROA*), and higher cashflow. Note that the difference in *MTB_Equity* is the only variable that is always statistically insignificant.

In Table 4, we examine *Premium1* across the high vs low analyst coverage subsamples. Consistent with our findings in Table 2 (and with Jory et al. (2016) who find lower M&A premium in the United States when either the acquirer or the target has a credit rating), our univariate test results show that targets with high analyst coverage tend to receive lower premium than those with low analyst coverage. The difference is always statistically significant at 5% level. Since target firms in emerging markets like China are more likely to be overvalued, the more negative *Premium1* measures in our high analyst coverage sample indicate faster and more immediate price correction. Our results in Table 4 are consistent with our first hypothesis.

Next, we use multivariate analysis to test whether targets with higher analyst coverage earn lower premium, by controlling for the firm characteristics. Our multivariate regression model is as follows:

[4] In the hypotheses section, we mentioned that targets may choose to sell a portion of the firm at seemingly lower price when: (1) Partial acquisition can enhance cooperation between two companies, (2) targets are facing financial constraints, (3) such acquisition can provide targets with access to acquirers' non-financial resources, or (4) acquirers can help provide certification effect to the targets to signal higher quality (Hertzel & Smith, 1993; Liao, 2013; Ouimet, 2013; and Fee et al., 2006).

Table 8
Robustness check based on acquisition methods.
In this table, the partial acquisition sample is further identified based on the different acquisition methods. Multivariate test is used for robustness checks. Results for negotiated transfer, asset stripping, and unrecognizable methods are presented in columns (1)–(3), (4)–(6), and (7)–(9), respectively. Only successful acquisitions are used. *T*-statistics are provided in parentheses. ***, **, and * denote significance at the 1%, 5% and 10% levels, respectively.

Control Variables	*Premium1*								
	Negotiated Transfer			Asset Stripping			Unrecognizable Methods		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
AnaDummy	−0.0493**			−0.0535**			−0.0422*		
	(-2.1159)			(-2.2129)			(-1.7574)		
AnaTimes		−0.0029**			−0.0016*			−0.0009**	
		(-2.3067)			(-2.0641)			(-2.5007)	
AnaNum			−0.0069**			−0.0037*			−0.0020**
			(-2.5358)			(-1.8941)			(-2.4959)
CashPay[9]	−0.1616	−0.1464	−0.1524				−0.1753	−0.2357***	−0.2344***
	(-1.2699)	(-0.9924)	(-1.0454)				(-1.6785)	(-17.4107)	(-17.8502)
Relevance	−0.0827**	−0.0845**	−0.0851**	−0.0209	−0.0110	−0.0028	0.3694*	0.3811	0.3858
	(-2.2242)	(-2.3451)	(-2.3651)	(-0.3467)	(-0.1302)	(-0.0321)	(1.9004)	(1.7565)	(1.7869)
MTB_Asset	−0.0327**	−0.0307*	−0.0301*	−0.0335	−0.0266	−0.0253	0.0016	−0.0022	−0.0021
	(-2.8158)	(-1.8442)	(-1.8092)	(-1.2271)	(-0.7098)	(-0.7432)	(0.2437)	(-0.7491)	(-0.6748)
MTB_Equity	0.0034	0.0036	0.0038	0.0231	0.0210	0.0308	0.0011	0.0040***	0.0041***
	(0.8343)	(0.7970)	(0.8498)	(0.9351)	(1.1419)	(1.2578)	(0.2698)	(3.7320)	(3.6529)
Growth	−0.0081	−0.0089	−0.0079	−0.0490	−0.0473	−0.0443	−0.0675***	−0.0627***	−0.0637***
	(-0.4264)	(-0.4637)	(-0.4049)	(-0.7791)	(-1.0355)	(-1.0661)	(-4.1018)	(-11.6977)	(-11.2803)
Leverage	−0.2378**	−0.2450***	−0.2530***	0.1073	0.1367	0.1018	0.0056	0.0200	0.0184
	(-2.2926)	(-2.7250)	(-2.8145)	(0.4637)	(1.3126)	(0.8124)	(0.0504)	(0.2889)	(0.2567)
LnAsset	0.0151	0.0197	0.0238	0.0028	0.0182	0.0216	0.0486***	0.0616***	0.0620***
	(0.8828)	(1.1024)	(1.2878)	(0.0625)	(0.3523)	(0.3976)	(3.7684)	(10.1319)	(10.0110)
Share	−0.2094***	−0.2235**	−0.2221**	−0.1147	−0.0521	−0.0152	−0.3205***	−0.3890***	−0.3890***
	(-3.5266)	(-2.5284)	(-2.5246)	(-0.9102)	(-0.8637)	(-0.3207)	(-3.7841)	(-7.6092)	(-7.5152)
ROA	−0.0776	−0.0605	−0.0531	0.7519	0.9725	0.8242	0.6153***	0.7845***	0.7947***
	(-0.2932)	(-0.3206)	(-0.2820)	(0.9876)	(1.4996)	(1.5463)	(3.6759)	(3.8526)	(3.8486)
OverConfidence	0.0467	0.0555	0.0439	0.0627	0.0603	0.0697	0.1975***	0.1989***	0.2026***
	(0.4746)	(0.6874)	(0.5531)	(0.5968)	(0.3947)	(0.4645)	(3.4901)	(4.8616)	(5.1310)
Cashflow	−0.0799	−0.0631	−0.0559	0.2002	0.0783	0.0824	−0.4758*	−0.6741	−0.6780
	(-0.5052)	(-0.3631)	(-0.3228)	(0.4787)	(0.1471)	(0.1337)	(-2.1012)	(-1.6605)	(-1.6816)
SDAR	−8.6810***	−8.8754***	−8.7837***	1.0470	1.6243	0.3049	0.1126	0.3148	0.2448
	(-4.4603)	(-4.0111)	(-3.9617)	(0.2253)	(0.5867)	(0.0830)	(0.0627)	(0.2805)	(0.2236)
CAR	0.1168**	0.1048	0.1071	0.7043***	0.7253***	0.7193***	0.0593	0.1011***	0.0984***
	(2.1421)	(1.3171)	(1.3545)	(3.6845)	(11.8214)	(10.0410)	(1.5675)	(4.9660)	(4.6399)
Intercept	0.0864	−0.0722	−0.1360	0.1203	−0.3154	−0.3825	−1.0799***	−1.2759**	−1.2555**
	(0.2384)	(-0.1610)	(-0.2965)	(0.1108)	(-0.2321)	(-0.2682)	(-3.8106)	(-3.2840)	(-3.1790)
N	598	598	598	137	137	137	318	318	318
adj.R^2	0.342	0.345	0.346	0.294	0.274	0.306	0.202	0.207	0.207

Table 9
Robustness Check with both Successful and Failed M&As.
This table presents multivariate results based on both successful and failed acquisitions. All control variables are provided in the left column. Three analyst coverage measures (*AnaDummy*, *AnaTimes*, *and AnaNum*) are used, while results are provided in columns (1), (2), and (3), respectively. All variables are winsorized at the 1 and 99 percentiles. *T*-statistics are provided in parentheses. ***, **, and * denote significance at the 1%, 5% and 10% levels, respectively.

Control Variables	Premium1		
	(1)	(2)	(3)
AnaDummy	−0.0363**		
	(-2.2464)		
AnaTimes		−0.0011**	
		(-2.0126)	
AnaNum			−0.0027**
			(-2.0828)
CashPay	−0.0206	−0.0237	−0.0245
	(-0.8003)	(-0.5983)	(-0.6020)
Relevance	−0.0882*	−0.0920***	−0.0912***
	(-1.7861)	(-3.1972)	(-3.0387)
MTB_Asset	0.0003	0.0011	0.0010
	(0.0419)	(0.1215)	(0.1163)
MTB_Equity	−0.0036	−0.0033	−0.0033
	(-1.6443)	(-1.3233)	(-1.3110)
Growth	−0.0222	−0.0233	−0.0237*
	(-1.6039)	(-1.6452)	(-1.6497)
Leverage	−0.1251	−0.1192**	−0.1215**
	(-1.2413)	(-2.2298)	(-2.2374)
LnAsset	0.0373*	0.0393***	0.0419***
	(2.1655)	(3.6178)	(3.6616)
Share	−0.1421*	−0.1582***	−0.1683***
	(-1.9846)	(-2.7739)	(-2.8863)
ROA	0.0719	0.0769	0.0872
	(0.7205)	(0.6510)	(0.7247)
OverConfidence	0.0532	0.0465	0.0447
	(0.7667)	(0.8301)	(0.7793)
Cashflow	−0.1361	−0.1416	−0.1376
	(-1.1206)	(-1.2212)	(-1.1695)
SDAR	−5.3226***	−5.3805***	−5.3331***
	(-3.9185)	(-3.8628)	(-3.7593)
CAR	0.1780***	0.1881***	0.1886***
	(3.4543)	(3.8082)	(3.6863)
Intercept	−0.8484*	−0.9052***	−0.9543***
	(-2.1019)	(-3.4425)	(-3.4648)
N	1260	1260	1260
adj.R^2	0.361	0.353	0.344

$$\begin{aligned}\text{Premium}_i &= \alpha + \beta_1\text{Analyst Coverage}_i + \beta_2\text{CashPay}_i + \beta_3\text{Relevance}_i + \beta_4\text{MTB\_Asset}_i + \beta_5\text{MTB\_Equity}_i + \\ &\beta_6\text{Growth}_i + \beta_7\text{Leverage}_i + \beta_8\text{LnAsset}_i + \beta_9\text{Share}_i + \beta_{10}\text{ROA}_i + \\ &\beta_{11}\text{OverConfidence}_i + \beta_{12}\text{Cashflow}_i + \beta_{13}\text{SDAR}_i + \beta_{14}\text{CAR}_i + \text{Industry}_i + \text{Year}_i + \varepsilon_i \qquad (1)\end{aligned}$$

Where the dependent variable, $Premium_i$ is the acquisition premium for target i. All three analyst coverage measures are used one at a time for robustness checks: *AnaDummy, AnaTimes,* and *AnaNum. Industry*$_i$ and *Year*$_i$ are dummy variables for target i, while the rest of the independent variables are defined and calculated in Appendix A.

Our multivariate results[5] in Table 5 are consistent with the univariate results in Table 4. Targets with high analyst coverage earn lower acquisition premium, after controlling for various firm characteristics. Results are statistically significant at 5% level. Therefore, higher analyst coverage brings overvalued stock to its equilibrium more quickly and more accurately. As a result, target firms with higher analyst coverage earn lower premium as a form of price correction. The control variables, if significant, carry the expected signs. For instance, the negative sign of a relative party acquisition (i.e., *Relevance* = 1) is consistent with Chen and Lu (2017), who find related mergers in China with lower M&A premiums.[6] While the negative coefficient of *MTB_Equity* is consistent with our first hypothesis that more overvalued targets will earn lower premium because of price correction, *MTB_Equity* is statistically significant only when *AnaDummy* is used as the analyst coverage measure in column (1). The negative growth in revenue (*Growth*) indicates that firms with higher growth tend to earn lower premium in China. Potentially, *Growth* captures a portion of the overvaluation in our analysis, since high

[5] To conserve space, we omit the *Year* and *Industry Dummy* in all of our tables from this point on.

[6] Reuter et al. (2012) find that the inter-organizational relationships of a firm matter. For a private firm, it can have significant impact on not only IPO pricing (Gulati & Higgins, 2003; Stuart, Hoang, & Hybels, 1999), but also on acquisition premiums.

Table 10
Adding proportion of the target acquisition to the model.
This table adds acquisition proportion to the multivariate analysis to examine the impact of analyst coverage on M&A premium. All control variables are provided in the left column. All variables are winsorized at the 1 and 99 percentiles. *T*-statistics are provided in parentheses. ***, **, and * denote significance at the 1%, 5% and 10% levels, respectively.

Control Variables	Premium1		
	(1)	(2)	(3)
AnaDummy	−0.0110		
	(-0.4790)		
AnaDummy × Proportion	−0.0735**		
	(-2.7075)		
AnaTimes		−0.0012*	
		(-1.6744)	
AnaTimes × Proportion		−0.0068***	
		(-3.8658)	
AnaNum			−0.0021*
			(-1.6543)
AnaNum × Proportion			−0.0105***
			(-2.6142)
Proportion	0.0065	−0.0006	−0.0018
	(0.2909)	(-0.0294)	(-0.0805)
CashPay	−0.0167	−0.0134	−0.0171
	(-0.5315)	(-0.3722)	(-0.4541)
Relevance	−0.0734**	−0.0703***	−0.0750***
	(-2.5386)	(-2.5856)	(-2.6641)
MTB_Asset	0.0017	0.0024	0.0027
	(0.4128)	(0.2942)	(0.3150)
MTB_Equity	−0.0040*	−0.0028	−0.0029
	(-1.8620)	(-1.1792)	(-1.2048)
Growth	−0.0291*	−0.0260*	−0.0255*
	(-1.9621)	(-1.9285)	(-1.7780)
Leverage	−0.0925	−0.1133**	−0.1163**
	(-1.4731)	(-2.2486)	(-2.2297)
LnAsset	0.0336***	0.0415***	0.0409***
	(3.5628)	(4.2670)	(4.0030)
Share	−0.1628**	−0.1297**	−0.1413***
	(-2.1431)	(-2.5009)	(-2.6138)
ROA	0.1158	0.1161	0.1210
	(0.6395)	(1.0462)	(1.0556)
OverConfidence	0.0556	0.0620	0.0499
	(0.8589)	(1.1859)	(0.9165)
Cashflow	−0.1527	−0.1249	−0.1435
	(-1.7095)	(-1.1747)	(-1.2990)
SDAR	−6.0007***	−5.2987***	−5.6228***
	(-3.3497)	(-4.2352)	(-4.2735)
CAR	0.1902***	0.1336***	0.1645***
	(5.8608)	(3.0589)	(3.5380)
Intercept	−0.7416***	−0.9816***	−0.9369***
	(-3.0822)	(-4.1358)	(-3.7653)
N	1207	1207	1207
adj.R^2	0.349	0.393	0.382

growth firms tend to have higher stock prices. Therefore, overvalued targets earn lower premium as price correction takes place. Please note that *CashPay* in Table 5 and in all of the following tables is almost always statistically insignificant. This is potentially due to the fact that 93.95% of our targets in partial acquisition sample are acquired with cash payment, as shown in Table 2.

To mitigate the potential endogeneity between acquisition premium and analyst coverage, we also use a two-stage least square regression analysis. We follow Yu (2008) and Xu et al. (2013) to calculate an expected analyst coverage (*E_Ana*) as follows:

$$E\_Ana_{i,k,t} = (Broker\_Ana_{k,t} / Broker\_Ana_{k,t-1}) \times AnaNum_{i,k,t-1} \quad (2)$$

Where $E\_Ana_{i,k,t}$ is the expected analyst coverage at brokerage firm k that follow target i in year t; $Broker\_Ana_{k,t}$ is the number of analysts at brokerage firm k in year t; and $AnaNum_{i,k,t-1}$ is target i's analyst coverage from broker k during [-720, −360] from the first acquisition announcement. Then, we calculate the expected analyst coverage for target i as follows:

$$E\_Ana_{i,t} = \sum_k E\_Ana_{i,k,t} \quad (3)$$

In the first stage of the two-stage regression, we use $E\_Ana_{i,t}$ as the instrumental variable in regression model (4) to estimate $AnaNum_{i,t}$, the number of analysts who follow target i at a given year.

Table 11
Heckman selection model for self-selection bias on M&A premium.
Heckman Selection Model is used to control for selection bias problem in this table to examine how analyst coverage influences M&A premium. $AnaCOV_{i,t}=1$ if the company i has analyst coverage in year t; otherwise, $AnaCOV_{i,t}=0$. $IndAnaCOV_{i,t}$ is the proportion of companies in the same industry as company i that have analyst coverage in year t. *T*-statistics are provided in parentheses. ***, **, and * denote significance at the 1%, 5% and 10% levels, respectively.

Control Variable	First Stage	Second Stage		
	AnaCOV		Premium1	
	(1)	(2)	(3)	(4)
AnaDummy		−0.0401**		
		(-1.9989)		
AnaTimes			−0.0013**	
			(-2.0169)	
AnaNum				−0.0028**
				(-1.9765)
IndAnaCOV	3.2295**			
	(14.0118)			
CashPay		−0.0160	−0.0183	−0.0202
		(-0.3889)	(-0.4796)	(-0.4966)
Relevance		−0.0875***	−0.0895***	−0.0903***
		(-2.9221)	(-3.2320)	(-2.9990)
MTB_Asset	0.1204***	−0.0023	−0.0009	−0.0006
	(9.3483)	(-0.2550)	(-0.1068)	(-0.0720)
MTB_Equity	−0.0065	−0.0029	−0.0026	−0.0027
	(-1.4250)	(-1.1859)	(-1.0581)	(-1.1015)
Growth	−0.0487***	−0.0233	−0.0239*	−0.0250*
	(-2.7786)	(-1.5507)	(-1.6596)	(-1.6603)
Leverage	−1.0194***	−0.0588	−0.0765	−0.0704
	(-8.8050)	(-0.9714)	(-1.3213)	(-1.1558)
LnAsset	0.6733***	0.0123	0.0208	0.0216
	(20.5291)	(0.8224)	(1.3916)	(1.3401)
Share	0.0227	−0.1675***	−0.1494***	−0.1756***
	(0.2246)	(−2.8766)	(−2.7164)	(−3.0176)
ROA	6.3596***	−0.2028	−0.0958	−0.1034
	(16.7034)	(-1.1680)	(-0.5612)	(-0.5755)
OverConfidence	−0.6624***	0.0819	0.0690	0.0656
	(-7.2758)	(1.3559)	(1.2069)	(1.0729)
Cashflow	0.6886***	−0.1885	−0.1897*	−0.1909
	(2.6615)	(-1.6036)	(-1.6952)	(-1.6217)
SDAR	12.9508***	−5.9008***	−5.6904***	−5.8412***
	(2.6431)	(-4.1945)	(-4.2541)	(-4.1224)
CAR	−0.0246	0.1837***	0.1747***	0.1881***
	(-0.4265)	(3.4912)	(3.6296)	(3.5742)
Lambda		−0.0864**	−0.0555	−0.0613*
		(-2.4275)	(-1.5827)	(-1.6773)
Intercept	−16.6365***	−0.0842	−0.3026	−0.2910
	(-21.0655)	(-0.2489)	(-0.8989)	(-0.8065)
N	24,052	1207	1207	1207
adj.R^2	0.302	0.349	0.371	0.348

$$AnaNum_{i,t}=\beta_0+\beta_1 E\_Ana_{i,t}+\gamma Control_{i,t}+Industry+Year+\varepsilon_{i,t} \tag{4}$$

We then use the estimated $AnaNum_{i,t}$ ($Pred\_AnaNum_{i,t}$) from stage one as the dependent variable in stage two of the two-stage least square regression analysis. The findings are presented in Table 6. The coefficient of $Pred\_AnaNum_i$ is negative and significant at the 5% level, which is consistent with the results in Table 5: Targets with higher analyst coverage earn lower premium as a form of price correction. In addition, targets in related acquisitions, high growth, and high price dispersion also earn lower premium, while larger targets and better performers earn higher premium.

In Table 7, we use our second premium measure, *Premium2*, for robustness checks. The coefficients of *AnaDummy, AnaTimes*, and *AnaNum* remain negatively significant at 5% level. The main findings in Table 7 remain consistent with our hypothesis.

In Table 8, we separate our targets into different subsamples based on their acquisition methods. We end up with 598 observations of negotiated acquisition, 137 observations of asset stripping, 318 observations of unrecognized methods, plus small samples of asset acquisition, debt payment, reorganization, equity investment, auction, and tender offer. Since most of the subsamples have very small sample sizes, we examine the three subsamples with sufficient sample sizes: negotiated transfer, asset stripping, and unrecognizable methods. In all three subsamples, the three analyst coverage measures remain significantly negative as before, while other control variables vary from one subsample to another. In Table 9, we perform further robustness checks by including both successful and failed partial acquisitions in our sample and find the three analyst coverage variables to remain significantly negative as before.

In Table 10, we examine whether the proportion of the targets acquired during the acquisition will affect acquirers' negotiation

Table 12
Adding corporate transparency to the model.
This table adds transparency to the multivariate analysis to examine the impact of analyst coverage on M&A premium. All control variables are provided in the left column. Three analyst coverage measures (*AnaDummy*, *AnaTimes, and AnaNum*) are used, while results are provided in columns (1), (2), and (3), respectively. Only successful acquisitions are used. All variables are winsorized at the 1 and 99 percentiles. *T*-statistics are provided in parentheses. ***, **, and * denote significance at the 1%, 5% and 10% levels, respectively.

Control Variables	Premium1		
	(1)	(2)	(3)
AnaDummy	−0.0978***		
	(-3.6192)		
AnaDummy × Transparency1	−0.3033***		
	(-3.6814)		
AnaTimes		−0.0031***	
		(-3.7358)	
AnaTimes × Transparency1		−0.0099***	
		(-3.7425)	
AnaNum			−0.0079***
			(-3.9683)
AnaNum × Transparency1			−0.0267***
			(-3.8574)
Transparency1	0.1876***	0.1098**	0.1324**
	(2.8634)	(2.0922)	(2.4485)
CashPay	−0.0242	−0.0213	−0.0209
	(-0.5868)	(-0.5249)	(-0.5150)
Relevance	−0.0848***	−0.0862***	−0.0865***
	(-2.8044)	(-2.8467)	(-2.8576)
MTB_Asset	0.0002	0.0012	0.0012
	(0.0252)	(0.1362)	(0.1351)
MTB_Equity	−0.0026	−0.0029	−0.0026
	(-1.0418)	(-1.1490)	(-1.0547)
Growth	−0.0293*	−0.0275*	−0.0282*
	(-1.9358)	(-1.8205)	(-1.8647)
Leverage	−0.1140**	−0.0993*	−0.1045*
	(-2.0025)	(-1.7808)	(-1.8711)
LnAsset	0.0344***	0.0363***	0.0376***
	(3.0624)	(3.2347)	(3.2863)
Share	−0.1782***	−0.1850***	−0.1848***
	(-3.0576)	(-3.1503)	(-3.1549)
ROA	0.0838	0.1042	0.0982
	(0.7139)	(0.8756)	(0.8261)
OverConfidence	0.0544	0.0529	0.0533
	(0.9031)	(0.8814)	(0.8865)
Cashflow	−0.1509	−0.1608	−0.1444
	(-1.2897)	(-1.3717)	(-1.2343)
SDAR	−5.5875***	−5.6119***	−5.6203***
	(-3.9351)	(-3.9397)	(-3.9441)
CAR	0.1882***	0.1925***	0.1905***
	(3.6085)	(3.6712)	(3.6409)
Intercept	−0.6966**	−0.7909***	−0.8134***
	(-2.5418)	(-2.9225)	(-2.9632)
N	1207	1207	1207
adj.R^2	0.351	0.351	0.352

power and therefore the acquisition premium. As shown in Panel C of Table 1, our targets on average sell only 10.41% of their shares during partial acquisitions. While we are unable to obtain shareholding ratio post acquisition, we are able to construct *Proportion* as follows:

Proportion is used to measure the portion of shares sold during the acquisition. If the proportion of the shares sold during the acquisition is greater than the average portion sold in the total sample, *Proportion* = 1; otherwise, *Proportion* = 0.

$$\text{Premium}_i = \alpha + \beta_1 \text{AnalystCoverage}_i + \beta_2\ \text{AnalystCoverage}_i \times \text{Proportion}_i + \beta_3 \text{Proportion}_i + \beta_4 \text{Control}_i + \text{Industry}_i + \text{Year}_i + \varepsilon_i \tag{5}$$

In Table 10, we see that while *Proportion* is always statistically insignificant, the interaction terms of *AnaDummy* × *Proportion, AnaTimes* × *Proportion, and AnaNum* × *Proportion* all have significantly negative relationship with *Premium*. This finding indicates that larger proportion of acquisition will magnify the impact of analyst coverage and further speed up price discovery.

Next, since existing literature shows that analysts prefer to cover firms with higher corporate transparency,[7] we use Heckman Selection Model (Busom, 2000; Heckman, 1979) to mitigate the selection bias problem. In stage one, all of the A-share listed companies in

[7] McNichols and O'Brien (1997).

Table 13
An alternative transparency measure.
This table uses an alternative transparency measure (*Transparency2*) in the multivariate analysis. All control variables are provided in the left column. Three analyst coverage measures (*AnaDummy*, *AnaTimes, and AnaNum*) are used, while results are provided in columns (1), (2), and (3), respectively. Only successful acquisitions are used. All variables are winsorized at the 1 and 99 percentiles. *T*-statistics are provided in parentheses. ***, **, and * denote significance at the 1%, 5% and 10% levels, respectively.

Control Variables	Premium1		
	(1)	(2)	(3)
AnaDummy	−0.0404**		
	(-2.0204)		
AnaDummy × Transparency2	−0.0662***		
	(-4.9578)		
AnaTimes		−0.0016***	
		(-2.6570)	
AnaTimes × Transparency2		−0.0063***	
		(-3.1611)	
AnaNum			−0.0035***
			(-2.6267)
AnaNum × Transparency2			−0.0107***
			(-3.2936)
Transparency2	0.1348	0.1456	0.1366
	(1.3559)	(1.4505)	(1.3708)
CashPay	−0.0209	−0.0232	−0.0229
	(-0.5084)	(-0.5707)	(-0.5642)
Relevance	−0.0851***	−0.0873***	−0.0878***
	(-2.8177)	(-2.8755)	(-2.8959)
MTB_Asset	0.0019	0.0023	0.0024
	(0.2106)	(0.2519)	(0.2699)
MTB_Equity	−0.0035	−0.0031	−0.0031
	(-1.4135)	(-1.2506)	(-1.2160)
Growth	−0.0297*	−0.0298*	−0.0297*
	(-1.9427)	(-1.9596)	(-1.9429)
Leverage	−0.1006*	−0.0980*	−0.1015*
	(-1.7653)	(-1.7487)	(-1.8109)
LnAsset	0.0354***	0.0371***	0.0386***
	(3.1243)	(3.2844)	(3.3522)
Share	−0.1763***	−0.1824***	−0.1818***
	(-3.0231)	(-3.1183)	(-3.1157)
ROA	0.0980	0.0966	0.1039
	(0.8134)	(0.7939)	(0.8522)
OverConfidence	0.0606	0.0573	0.0533
	(1.0255)	(0.9620)	(0.8932)
Cashflow	−0.1675	−0.1695	−0.1664
	(-1.4248)	(-1.4387)	(-1.4144)
SDAR	−5.8311***	−5.8278***	−5.8074***
	(-4.1085)	(-4.0895)	(-4.0724)
CAR	0.1880***	0.1918***	0.1907***
	(3.5675)	(3.6436)	(3.6253)
Intercept	−0.5643**	−0.6175**	−0.6431**
	(-2.0932)	(-2.2555)	(-2.3110)
N	1207	1207	1207
adj.R^2	0.347	0.348	0.348

the 2004–2016 sample period are used. We use Probit model in equation (6) to estimate the probability that the target company will have analyst coverage. Then, we recover the Inverse Mill's Ratio (*Lambda*) from stage one and put it into stage two.

$$\mathrm{AnaCOV}_{i,t} = \beta_0 + \beta_1 \mathrm{IndAnaCOV}_{i,t} + \gamma \mathrm{OtherVariables}_{i,t} + \varepsilon_{i,t} \tag{6}$$

Where $AnaCOV_{i,t} = 1$ if the company i has analyst coverage in year t; otherwise, $AnaCOV_{i,t} = 0$. $IndAnaCOV_{i,t}$ is the proportion of companies in the same industry as company i that have analyst coverage in year t. *OtherVariable* is a list of other control variables used in our analysis. Please note that all listed companies (including non-M&A firms) are used in the first stage of Heckman Selection Model. Since *CashPay* and *Relevance* are dummy variables which only apply to M&A firms, these variables are omitted in stage one of the Heckman Selection Model.

In Table 11, we can see that *Lambda* is always negative, although only statistically significant in two out of three regressions. Our results are consistent with the literature and confirm the existence of selection bias in our sample. However, after controlling for selection bias with the Heckman Selection Model, *AnaDummy*, *AnaTimes*, and *AnaNum* remain significantly negative with acquisition premium and consistent with our hypothesis. Next, we incorporate transparency directly in Tables 12 and 13 since less transparent firms tend to be more mispriced (Beatty & Ritter, 1986), while the lack of transparency is one of the major concerns for investors in the emerging markets. We first follow Hutton, Marcus, and Tehranian (2009) and calculate *Transparency1* by using three years of data (t-3 to

Table 14
The impact of analyst forecast revision on M&A premium.
This table adds analyst forecast revision to the multivariate analysis. All control variables are provided in the left column. Since only targets with analyst following are used, *AnaDummy* is excluded from the analysis. Results based on *AnaTimes and AnaNum* are provided in columns (1) and (2), respectively. Only successful acquisitions are used. All variables are winsorized at the 1 and 99 percentiles. *T*-statistics are provided in parentheses. ***, **, and * denote significance at the 1%, 5% and 10% levels, respectively.

Control Variables	Premium1	
	(1)	(2)
AnaTimes	−0.0015**	
	(-2.0236)	
AnaTimes × Adjust	−0.0964**	
	(-2.5355)	
AnaNum		−0.0035**
		(-2.0217)
AnaNum × Adjust		−0.2197***
		(-2.8578)
Adjust	1.8998	2.2160*
	(1.5946)	(1.8404)
CashPay	−0.0935	−0.0948
	(-0.9380)	(-0.9545)
Relevance	−0.1800***	−0.1798***
	(-4.1003)	(-4.1032)
MTB_Asset	−0.0143	−0.0150
	(-0.7081)	(-0.7402)
MTB_Equity	0.0108	0.0111
	(0.9576)	(0.9817)
Growth	−0.0406	−0.0410
	(-1.5047)	(-1.5379)
Leverage	0.0856	0.0764
	(0.7281)	(0.6503)
LnAsset	0.0576***	0.0571***
	(3.1439)	(3.0478)
Share	−0.2283**	−0.2281**
	(-2.5514)	(-2.5458)
ROA	0.6838*	0.6762
	(1.6683)	(1.6365)
OverConfidence	0.2893**	0.2797**
	(2.4379)	(2.3307)
Cashflow	−0.2354	−0.2196
	(-0.9763)	(-0.9144)
SDAR	−2.6403	−2.6560
	(-1.1530)	(-1.1618)
CAR	0.2243***	0.2266***
	(2.8059)	(2.8463)
Intercept	−1.3136***	−1.2833***
	(-2.9707)	(-2.8578)
N	474	474
adj.R^2	0.229	0.229

t-1) from the announcement date. However, corporate transparency is a very complex issue in this case. If high transparency firms are in general better performers and have higher profitability (Singhvi & Desai, 1971; Lang & Lundholm, 1993; and Healy, Hutton, & Palepu, 1999), high transparency firms should earn higher acquisition premium. However, if higher transparency helps analyst coverage to further speed up price discovery, our overvaluation hypothesis predicts a negative relationship with acquisition premium.

In Table 12, *Transparency1* seems to capture both sides of the story and show that higher transparency helps acquirers identify good performers from the mispriced targets in our sample. Therefore, high transparency firms earn higher premium. Next, all of the analyst coverage coefficients are now more statistically significant than before at 1% level, while the sizes of the three analyst coverage coefficients have more than doubled from Table 5. The negative interactions terms of *AnaDummy* × *Transparency1, AnaTimes* × *Transparency1* and *AnaNum* × *Transparency1* remain consistent with our hypothesis that higher transparency combined with higher analyst coverage can further speed up price discovery. As a result, overvalued targets are acquired at lower premium as a form of price correction.

Note that *Transparency1* in Table 12 is calculated using the "three-year" data before the announcement date, while corporations can change their voluntary disclosure and therefore transparency from time to time. Therefore, we also calculate an alternative transparency measure, *Transparency2*, using only the transparency data in the year before the announcement for hopefully a more precise transparency measure and more reliable results. In Table 13, we see that while *Transparency2* by itself is always statistically insignificant, all analyst coverage variables and all interaction terms with *Transparency2* are significantly negative. Therefore, while higher corporate transparency in the year before acquisition does not seem to impact acquisition premium by itself, it does help analysts make better and

Table 15
The impact of analyst forecast precision on M&A premium.
This table adds analyst forecast precision to the multivariate analysis. All control variables are provided in the left column. Since only targets with analyst following are used, *AnaDummy* is excluded from the analysis. Results based on *AnaTimes and AnaNum* are provided in columns (1) and (2), respectively. Only successful acquisitions are used. All variables are winsorized at the 1 and 99 percentiles. *T*-statistics are provided in parentheses. ***, **, and * denote significance at the 1%, 5% and 10% levels, respectively.

Control Variables	Premium1	
	(1)	(2)
AnaTimes	0.0013	
	(1.5394)	
AnaTimes × Precision	−0.0032**	
	(-2.5758)	
AnaNum		0.0027
		(1.0280)
AnaNum × Precision		−0.0067**
		(-2.0828)
Precision	0.0229	0.0257
	(0.6706)	(0.6147)
CashPay	−0.0697	−0.0691
	(-1.1423)	(-0.8322)
Relevance	−0.1736***	−0.1731***
	(-3.9183)	(-4.2446)
MTB_Asset	−0.0205	−0.0198
	(-1.3085)	(-1.1154)
MTB_Equity	0.0122	0.0119
	(1.1689)	(1.1687)
Growth	−0.0722***	−0.0726***
	(-4.0080)	(-3.1423)
Leverage	−0.0177	−0.0190
	(-0.1225)	(-0.1713)
LnAsset	0.0435**	0.0431**
	(2.3176)	(2.4641)
Share	−0.2489*	−0.2489**
	(-2.0503)	(-2.5689)
ROA	0.4912	0.4719
	(1.5787)	(1.4422)
OverConfidence	0.2269*	0.2171**
	(2.0111)	(2.0667)
Cashflow	−0.1660	−0.1579
	(-0.9631)	(-0.7308)
SDAR	−3.7746*	−3.9144*
	(-1.8844)	(-1.9290)
CAR	0.2302***	0.2281***
	(5.1702)	(2.9995)
Intercept	−1.0585**	−0.9147**
	(-2.5781)	(-2.1095)
N	574	574
adj.R^2	0.271	0.269

more informative decisions. Therefore, higher transparency, combined with analyst coverage, can promote market efficiency and speed up price correction.

In Table 14, we examine whether analyst forecast revisions also help to speed up price discovery in the acquisition market. We follow Chan, Jegadeesh, and Lakonishok (1996), Gleason and Lee (2003), and Jegadeesh, Kim, Krische, and Lee (2004) to calculate *Adjust* as the average of the analyst forecast revisions, while details of the calculation is provided in Appendix A. *AnaTime* and *AnaNum* remain significantly negative, while the negative interaction terms of *AnaTime* × *Adjust* and *AnaNum* × *Adjust* indicate that new and useful rather than old information is released to the market when analysts provide forecast revisions. Therefore, analyst forecast revisions also speed up price discovery and reduce acquisition premium.

In Table 15, since studies have found analyst forecast quality to be questionable at times when there are personal interests involved[8], we also incorporate analyst forecast precision in the analysis by following Duru and Reeb (2002). *AnaDummy* is excluded from the analysis since firms must have analyst coverage before *Precision* can be constructed. We find that *Precision* by itself does not explain acquisition premium. However, the interactions of *AnaTimes* × *Precision* and *AnaNum* × *Precision* are both significantly negative, which indicate that analyst forecast precision combined with higher analyst coverage can speed up price correction and improve market efficiency.

[8] Michaely & Womack, 1999; Jackson, 2005; McNichols, O'Brian, and Pamukcu, 2007; and Demiroglu & Ryngaert, 2010.

[9] Note that since all asset stripping events are paid with cash payment, *CashPay* is left blank in columns (4), (5), and (6).

Table 16

Star analysts versus non-star analysts on M&A premium.

This table resents Chi-square test results on star analysts versus non-star analysts groups. Three analyst coverage measures (*AnaDummy*, *AnaTimes, and AnaNum*) are used, while results are provided in columns (1)–(2), (3)–(4), and (5)–(6), respectively. Analyst coverage measures initiated with "Com" represent non-star analyst measures, while analyst coverage measures initiated with "Star" represent star analyst measures. All variables are winsorized at the 1 and 99 percentiles. *T*-statistics are provided in parentheses, while Chi-square results are provided in the bottom row. ***, **, and * denote significance at the 1%, 5% and 10% levels, respectively.

Control Variables	Premium1					
	(1)	(2)	(3)	(4)	(5)	(6)
ComAnaDummy	−0.0334*					
	(-1.6870)					
StarAnaDummy		−0.0574*				
		(-1.8821)				
ComAnaTimes			−0.0011*			
			(-1.9136)			
StarAnaTimes				−0.0201***		
				(-3.3055)		
ComAnaNum					−0.0027**	
					(-2.0560)	
StarAnaNum						−0.0542**
						(-2.2307)
CashPay	−0.0194	−0.0227	−0.0221	−0.0221	−0.0219	−0.0225
	(-0.4817)	(-0.5556)	(-0.5416)	(-0.8748)	(-0.5376)	(-0.5502)
Relevance	−0.0862***	−0.0877***	−0.0884***	−0.0892	−0.0888***	−0.0885***
	(-2.9756)	(-2.9197)	(-2.9286)	(-1.6357)	(-2.9407)	(-2.9460)
MTB_Asset	0.0019	0.0015	0.0021	0.0018	0.0024	0.0016
	(0.2148)	(0.1644)	(0.2374)	(0.2425)	(0.2614)	(0.1736)
MTB_Equity	−0.0038	−0.0038	−0.0035	−0.0038	−0.0034	−0.0038
	(-1.5362)	(-1.5021)	(-1.3888)	(-1.5455)	(-1.3510)	(-1.4922)
Growth	−0.0274*	−0.0299**	−0.0283*	−0.0297*	−0.0280*	−0.0301**
	(-1.8833)	(-2.0328)	(-1.9194)	(-1.9053)	(-1.8914)	(-2.0452)
Leverage	−0.0977*	−0.0882	−0.0945*	−0.0901	−0.0982*	−0.0882
	(-1.7392)	(-1.5730)	(-1.6883)	(-0.9151)	(-1.7540)	(-1.5739)
LnAsset	0.0339***	0.0348***	0.0369***	0.0355**	0.0386***	0.0358***
	(3.0918)	(3.2034)	(3.2693)	(2.3337)	(3.3542)	(3.2935)
Share	−0.1626***	−0.1811***	−0.1763***	−0.1869**	−0.1767***	−0.1796***
	(-2.8577)	(-3.1143)	(-3.0210)	(-2.6560)	(-3.0332)	(-3.0960)
ROA	0.1083	0.1015	0.1168	0.0926	0.1234	0.0977
	(0.9183)	(0.8467)	(0.9652)	(0.9638)	(1.0177)	(0.8143)
OverConfidence	0.0527	0.0518	0.0481	0.0508	0.0446	0.0521
	(0.9128)	(0.8728)	(0.8087)	(0.5587)	(0.7484)	(0.8781)
Cashflow	−0.1668	−0.1753	−0.1718	−0.1732	−0.1696	−0.1688
	(-1.4399)	(-1.4826)	(-1.4575)	(-1.7237)	(-1.4428)	(-1.4239)
SDAR	−5.8149***	−5.7454***	−5.7581***	−5.7615***	−5.7356***	−5.7430***
	(-4.1744)	(-4.0466)	(-4.0468)	(-3.4949)	(-4.0290)	(-4.0385)
CAR	0.1897***	0.1969***	0.1929***	0.1986***	0.1921***	0.2003***
	(3.7073)	(3.7115)	(3.6555)	(3.1598)	(3.6459)	(3.7720)
Intercept	−0.7471***	−0.7807***	−0.8232***	−0.7979**	−0.8561***	−0.8028***
	(-2.8072)	(-2.9566)	(-3.0209)	(-2.2221)	(-3.0864)	(-3.0384)
N	1207	1207	1207	1207	1207	1207
adj.R^2	0.355	0.347	0.347	0.348	0.347	0.348
Chi-square Test for Non-Star Analysts minus Star Analysts	$\chi^2 = 3.15$*		$\chi^2 = 11.89$***		$\chi^2 = 5.04$**	
	(p = 0.0758)		(p = 0.0006)		(p = 0.0248)	

Next, some may argue that star analysts can bring reputational or halo effect to the firm without actually providing valuable information (Brown & Perry, 1994), while others find star analysts to outperform benchmarks (Desai, Liang, & Singh, 2000) and provide more accurate earnings forecasts than non-star analysts (Xu et al., 2013). Therefore, we obtain the variable from the CSMAR and divide our sample into star analyst and non-star analyst subsamples in Table 16 to examine their impact on acquisition premium. *ComAnaDummy* = 1 if the firm is followed by a non-star analyst; otherwise, *ComAnaDummy* = 0. *ComAnaTimes* (*StarAnaTimes*) is the number of times the firms is followed by any non-start analysts (star analysts). *ComAnaNum* (*StarAnaNum*) is the number of non-star (star) analysts who follow the firm. *StarAnaDummy* = 1 if the firm is followed by a star analyst; otherwise, *StarAnaDummy* = 0.

In Table 16, we find that while both non-star analysts and star analysts can effectively speed up price correction, targets with star analysts earn lower premium than those with non-star analysts. Chi-square test on the difference is always statistically significant. Our results indicate that while both star and non-star analysts provide valuable information to external investors and promote market efficiency, star analysts are more effective at doing so.

To make sure our finding of negative correlation between analyst coverage and acquisition premium is the result of price

Table 17
The impact of analyst coverage on EPS.
This table presents multivariate results on how analyst coverage affects EPS of the targets. All control variables are provided in the left column. Three analyst coverage measures (*AnaDummy*, *AnaTimes*, *and AnaNum*) are used. Results for EPS in years t, t+1, and t+2 are provided in columns (1)–(3), (4)–(6), and (7)–(9), respectively. Only successful acquisitions are used. All variables are winsorized at the 1 and 99 percentiles. *T*-statistics are provided in parentheses. ***, **, and * denote significance at the 1%, 5% and 10% levels, respectively.

Control Variables	EPS_t			EPS_{t+1}			EPS_{t+2}		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
AnaDummy	0.1407***			0.1310***			0.1346***		
	(4.7329)			(4.3306)			(3.9923)		
AnaTimes		0.0073***			0.0063***			0.0060***	
		(7.1376)			(5.1903)			(3.5567)	
AnaNum			0.0165***			0.0146***			0.0146***
			(7.3403)			(5.6005)			(4.0667)
CashPay	0.0370	0.0417	0.0417	0.0549	0.0605	0.0601	−0.0202	−0.0124	−0.0137
	(0.5772)	(0.6584)	(0.6638)	(0.8788)	(0.9931)	(0.9912)	(-0.2280)	(-0.1427)	(-0.1581)
Relevance	0.0394	0.0568	0.0582	0.0037	0.0193	0.0208	0.0423	0.0633	0.0650
	(0.8851)	(1.2988)	(1.3284)	(0.0733)	(0.3850)	(0.4139)	(0.6984)	(1.0569)	(1.0884)
MTB_Asset	0.0388**	0.0348**	0.0339**	0.0246	0.0210	0.0201	−0.0163	−0.0218	−0.0241
	(2.4798)	(2.2665)	(2.2125)	(1.5922)	(1.3812)	(1.3286)	(-1.0015)	(-1.4170)	(-1.5690)
MTB_Equity	−0.0062	−0.0091	−0.0095	−0.0051	−0.0076	−0.0080	0.0091*	0.0074	0.0070
	(-0.8364)	(-1.2342)	(-1.2889)	(-0.9550)	(-1.3669)	(-1.4429)	(1.7264)	(1.4300)	(1.3526)
Growth	0.0558**	0.0525**	0.0518**	0.0334	0.0314	0.0304	0.0247	0.0268	0.0253
	(2.2601)	(2.1114)	(2.0731)	(1.4054)	(1.2959)	(1.2514)	(0.7758)	(0.8207)	(0.7703)
Leverage	−0.1156	−0.0891	−0.0729	0.0674	0.0880	0.1042	−0.0164	0.0047	0.0235
	(-1.1373)	(-0.8788)	(-0.7202)	(0.7085)	(0.9205)	(1.0856)	(-0.1413)	(0.0398)	(0.1999)
LnAsset	0.1361***	0.1074***	0.1004***	0.1038***	0.0807***	0.0731***	0.1370***	0.1150***	0.1046***
	(6.6010)	(5.2969)	(4.9298)	(4.9024)	(3.9096)	(3.5266)	(4.8778)	(4.3416)	(3.9458)
Share	−0.0163	0.0205	0.0187	−0.0479	−0.0146	−0.0156	−0.1696	−0.1349	−0.1302
	(-0.1551)	(0.2025)	(0.1851)	(-0.4357)	(-0.1360)	(-0.1458)	(-1.2822)	(-1.0472)	(-1.0120)
ROA	1.6103***	1.5024***	1.4739***	1.0506***	0.9684***	0.9370***	0.8783***	0.8342***	0.7978***
	(5.5528)	(5.2317)	(5.1417)	(4.6135)	(4.3659)	(4.2333)	(2.8697)	(2.7039)	(2.5962)
OverConfidence	0.1078	0.1453*	0.1647*	0.0815	0.1115	0.1299	0.1740	0.1962*	0.2169**
	(1.2045)	(1.6818)	(1.8877)	(0.8637)	(1.2035)	(1.4023)	(1.5681)	(1.8039)	(2.0016)
Cashflow	1.1994***	1.1958***	1.1875***	1.1404***	1.1473***	1.1370***	0.8406***	0.8541***	0.8408***
	(5.8171)	(5.8745)	(5.8410)	(5.8687)	(5.9398)	(5.8849)	(3.0369)	(3.1090)	(3.0620)
SDAR	3.0345	2.5008	2.4479	3.1277	2.6689	2.5978	12.8220***	12.4956***	12.2846***
	(1.1950)	(1.0076)	(0.9929)	(1.1410)	(0.9951)	(0.9761)	(3.5057)	(3.4888)	(3.4643)
CAR	0.3908***	0.3750***	0.3794***	0.4737***	0.4580***	0.4620***	0.2612**	0.2440*	0.2506*
	(4.5024)	(4.3985)	(4.4399)	(4.8068)	(4.7158)	(4.7588)	(1.9850)	(1.8853)	(1.9355)
Intercept	−2.9666***	−2.2673***	−2.1393***	−2.8754***	−2.3891***	−2.2859***	−2.9667***	−2.4106***	−2.2060***
	(-6.0408)	(-4.7248)	(-4.4562)	(-5.3971)	(-4.7804)	(-4.6191)	(-4.6056)	(-3.9795)	(-3.6620)
N	1207	1207	1207	1204	1204	1204	1030	1030	1030
adj.R^2	0.341	0.363	0.364	0.234	0.249	0.252	0.177	0.188	0.192

Table 18
Analyst coverage and EPS: 2SLS regression.
This table presents 2-stage least square regression results of the effects of analyst coverage on EPS. Only successful acquisitions are used. In the first stage of the two-stage regression, $E\_Ana_{i,t}$ is used as the instrumental variable in regression model (4) to estimate $AnaNum_{i,t}$, the number of analysts who follow target i at a given year. The estimated $AnaNum_{i,t}$ ($Pred\_AnaNum_{i,t}$) from stage one is then used as the dependent variable in stage two of the two-stage least square regression analysis. Results for EPS in years t, t+1, and t+2 are provided in columns (1)–(2), (3)–(4), and (5)–(6), respectively. *T*-statistics are provided in parentheses. ***, **, and * denote significance at the 1%, 5% and 10% levels, respectively.

Control Variables	EPS_t		EPS_{t+1}		EPS_{t+2}	
	Stage 1	Stage 2	Stage 1	Stage 2	Stage 1	Stage 2
	AnaNum (1)	EPS_t (2)	AnaNum (3)	EPS_{t+1} (4)	AnaNum (5)	EPS_{t+2} (6)
AnaNum		0.0150***		0.0137***		0.0164***
		(7.3088)		(5.4801)		(3.7218)
CashPay	0.0417	0.0430	0.0596	0.0608	−0.0164	−0.0152
	(0.7112)	(0.7694)	(0.8771)	(0.9566)	(-0.1884)	(-0.1794)
Relevance	0.0520	0.0570*	0.0154	0.0201	0.0638	0.0668
	(1.3977)	(1.6496)	(0.4518)	(0.6725)	(1.0774)	(1.1485)
MTB_Asset	0.0341	0.0346	0.0202	0.0206	−0.0272*	−0.0256*
	(1.4548)	(1.5686)	(1.1886)	(1.3566)	(-1.7459)	(-1.7056)
MTB_Equity	−0.0078	−0.0091	−0.0066	−0.0077	0.0076	0.0067
	(-0.7071)	(-0.9069)	(-0.9226)	(-1.1572)	(1.4760)	(1.3042)
Growth	0.0640	0.0530	0.0411	0.0311	0.0344	0.0243
	(1.7740)	(1.5638)	(1.6208)	(1.2609)	(1.0622)	(0.7499)
Leverage	−0.1048	−0.0827	0.0768	0.0982	0.0076	0.0363
	(-1.3103)	(-1.1118)	(0.6591)	(0.8971)	(0.0645)	(0.3088)
LnAsset	0.1180***	0.1061***	0.0876***	0.0765***	0.1097***	0.0973***
	(8.3568)	(8.9548)	(4.0446)	(3.4672)	(4.0541)	(3.4975)
Share	0.0260	0.0159	−0.0088	−0.0173	−0.1175	−0.1254
	(0.5539)	(0.3690)	(-0.1344)	(-0.2868)	(-0.9185)	(-1.0020)
ROA	1.5779***	1.4982***	1.0234**	0.9509***	0.8610***	0.7734**
	(3.8179)	(3.9025)	(3.0098)	(3.1667)	(2.7995)	(2.5749)
OverConfidence	0.1377	0.1599**	0.1084	0.1275	0.2120*	0.2213**
	(1.6279)	(2.1915)	(0.9000)	(1.1236)	(1.9286)	(2.0916)
Cashflow	1.2172***	1.1971***	1.1590***	1.1421***	0.8552***	0.8308***
	(5.7159)	(5.8984)	(7.0047)	(7.1116)	(3.1002)	(3.0798)
SDAR	3.9785	2.5713	3.9908	2.6769	14.1077***	12.1332***
	(0.9339)	(0.7057)	(0.7872)	(0.6012)	(3.9920)	(3.4900)
CAR	0.3907***	0.3800***	0.4732***	0.4625***	0.2630**	0.2500**
	(4.5458)	(4.7892)	(5.8237)	(6.1869)	(2.0209)	(1.9782)
E_Ana	0.0124***		0.0113***		0.0139***	
	(7.1050)		(5.2852)		(3.6213)	
Intercept	−2.5202***	−2.4027***	−2.5848***	−1.9067***	−2.3339***	−2.3182***
	(-7.3602)	(-8.7930)	(-5.0955)	(-3.4584)	(-3.8035)	(-3.5963)
N	1207	1207	1204	1204	1030	1030
adj.R^2	0.348	0.364	0.240	0.252	0.189	0.192

correction rather than price discount for poor performers, we examine EPS post successful acquisition. Recall from Panel C of Table 1, on average, 10.41% of each target firm is acquired at a time in our sample, while partial acquisitions allow the targets to remain as public firms post acquisition. Therefore, this partial acquisition sample allows us to examine the performance of the target firms post successful acquisition. In Table 17, the positive *AnaDummy*, *AnaTimes*, and *AnaNum* show that the lower acquisition premium of targets with higher analyst coverage is not the result of poor performance. Consistent with our hypotheses, it is the reduction of information asymmetry that allows the acquirers make better investment decisions. In this case, the acquirers can more accurately identify suitable targets and estimate their fair market value. Therefore, targets with higher analyst coverage tend to perform better in the long run, as indicated by higher EPS in year t to year t+2. Our results are significant at 1% level. This finding is consistent with Lang and Lundholm (1993) who find firms with higher transparency to have better firm performance and with Jory et al. (2016) who find merged firms with better operating performance post acquisition when the targets have credit ratings before the acquisition.

To mitigate the potential endogeneity problem and self-selection bias between analyst coverage and EPS, we also apply two-stage least square regression analysis and Heckman Selection Model in Tables 18 and 19, respectively. Our results remain consistent with our second hypothesis.

To provide a more comprehensive study, we also look at the relationship between acquisition premium and future EPS of the targets in Table 20, while both premium measures are used for robustness checks. We find that when the premium measure is statistically significant in Table 18, the negative coefficient is consistent with our price correction hypothesis. Targets with lower acquisition premium perform better in the long run post acquisition. Therefore, the lower premium is not the result of price discount for poor performers, but price correction to bring the overvalued stock prices closer to their fair market values. In addition, the negative premium coefficient is also consistent with Roll (1986) and Moeller, Schlingemann, and Stulz (2004): Overconfident managers are more likely to

Table 19
Heckman selection model for self-selection bias on EPS.
This table uses Heckman Selection Model to control for selection bias problem. In this table, we examine how analyst coverage influences EPS of the targets. $AnaCOV_{i,t} = 1$ if the company i has analyst coverage in year t; otherwise, $AnaCOV_{i,t} = 0$. $IndAnaCOV_{i,t}$ is the proportion of companies in the same industry as company i that have analyst coverage in year t. Results for EPS in years t, t+1, and t+2 are presented in columns (2)–(4), (5)–(7), and (8)–(10), respectively. *T*-statistics are provided in parentheses. ***, **, and * denote significance at the 1%, 5% and 10% levels, respectively.

Control Variables	First Stage	Second Stage								
	AnaCOV		EPS_t			EPS_{t+1}			EPS_{t+2}	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
AnaDummy		0.1444***			0.1379***			0.1415***		
		(4.8502)			(4.5791)			(4.2332)		
AnaTimes			0.0076***			0.0062***			0.0060***	
			(7.0446)			(4.9029)			(3.3129)	
AnaNum				0.0173***			0.0145***			0.0147***
				(7.2099)			(5.3192)			(3.8298)
IndAnaCOV	3.2295***									
	(14.0118)									
CashPay		0.0335	0.0436	0.0437	0.0490	0.0601	0.0599	−0.0274	−0.0123	−0.0132
		(0.5201)	(0.6874)	(0.6948)	(0.7874)	(0.9834)	(0.9841)	(-0.3104)	(-0.1407)	(-0.1511)
Relevance		0.0419	0.0551	0.0564	0.0080	0.0197	0.0210	0.0477	0.0632	0.0645
		(0.9435)	(1.2480)	(1.2778)	(0.1590)	(0.3908)	(0.4165)	(0.7858)	(1.0431)	(1.0685)
MTB_Asset	0.1204***	0.0430***	0.0310**	0.0299*	0.0315**	0.0219	0.0206	−0.0098	−0.0220	−0.0249
	(9.3483)	(2.6334)	(1.9973)	(1.9328)	(1.9728)	(1.3869)	(1.3164)	(-0.5723)	(-1.3275)	(-1.5070)
MTB_Equity	−0.0065	−0.0072	−0.0085	−0.0088	−0.0068	−0.0077	−0.0080	0.0073	0.0075	0.0072
	(-1.4250)	(-0.9455)	(-1.1616)	(-1.2158)	(-1.2525)	(-1.3834)	(-1.4478)	(1.3675)	(1.4339)	(1.3782)
Growth	−0.0487***	0.0511**	0.0557**	0.0550**	0.0255	0.0307	0.0300	0.0177	0.0270	0.0259
	(-2.7786)	(1.9939)	(2.1899)	(2.1567)	(1.0347)	(1.2454)	(1.2134)	(0.5321)	(0.8098)	(0.7754)
Leverage	−1.0194***	−0.1554	−0.0506	−0.0323	0.0012	0.0793	0.0989	−0.0795	0.0066	0.0319
	(-8.8050)	(-1.4677)	(-0.4798)	(-0.3055)	(0.0114)	(0.7499)	(0.9306)	(-0.6542)	(0.0499)	(0.2424)
LnAsset	0.6733***	0.1599***	0.0845***	0.0763**	0.1443***	0.0859***	0.0763**	0.1746***	0.1138***	0.0997**
	(20.5291)	(5.8499)	(2.8783)	(2.5767)	(4.8042)	(2.6751)	(2.3651)	(4.3775)	(2.5810)	(2.2538)
Share	0.0227	−0.0196	0.0246	0.0228	−0.0526	−0.0154	−0.0161	−0.1707	−0.1348	−0.1296
	(0.2246)	(-0.1868)	(0.2420)	(0.2251)	(-0.4817)	(-0.1441)	(-0.1506)	(-1.2935)	(-1.0453)	(-1.0068)
ROA	6.3596***	1.9365***	1.2173***	1.1764***	1.6097***	1.0338***	0.9766**	1.4080***	0.8204	0.7371
	(16.7034)	(5.2156)	(3.0767)	(2.9783)	(4.4005)	(2.6405)	(2.4991)	(3.0375)	(1.5715)	(1.4167)
OverConfidence	−0.6624***	0.0795	0.1702*	0.1914**	0.0311	0.1057	0.1263	0.1228	0.1975*	0.2228*
	(-7.2758)	(0.8617)	(1.8774)	(2.0834)	(0.3227)	(1.0975)	(1.3058)	(1.0614)	(1.6748)	(1.8877)
Cashflow	0.6886***	1.2264***	1.1671***	1.1574***	1.1889***	1.1540***	1.1411***	0.8911***	0.8526***	0.8339***
	(2.6615)	(5.8837)	(5.6182)	(5.5759)	(6.0774)	(5.8837)	(5.8145)	(3.2098)	(3.0505)	(2.9873)
SDAR	12.9508***	3.1821	2.3049	2.2435	3.3390	2.7111	2.6233	13.0842***	12.4853***	12.2385***
	(2.6431)	(1.2506)	(0.9223)	(0.9034)	(1.2205)	(1.0034)	(0.9782)	(3.5680)	(3.4408)	(3.4095)
CAR	−0.0246	0.3974***	0.3691***	0.3735***	0.4840***	0.4593***	0.4627***	0.2731**	0.2437*	0.2492*
	(-0.4265)	(4.6068)	(4.3797)	(4.4192)	(4.9306)	(4.7314)	(4.7700)	(2.1036)	(1.9050)	(1.9457)
Lambda		0.0891	−0.0743	−0.0773	0.1525**	0.0170	0.0103	0.1414	−0.0035	−0.0154
		(1.3209)	(-1.0358)	(-1.0794)	(2.0834)	(0.2186)	(0.1325)	(1.6410)	(-0.0361)	(-0.1584)
Intercept	−16.6365***	−3.7624***	−2.1337***	−1.5680**	−3.7508***	−1.8292**	−1.6514**	−3.8260***	−2.3575**	−2.0753**
	(-21.0655)	(-6.2987)	(-3.3389)	(-2.2274)	(-5.2678)	(-2.3285)	(-2.1038)	(-3.9304)	(-2.2358)	(-1.9729)
N	24,052	1207	1207	1207	1204	1204	1204	1030	1030	1030
adj.R^2	0.302	0.341	0.363	0.364	0.236	0.249	0.252	0.179	0.187	0.191

Table 20
The correlation between acquisition premium and EPS.
This table presents the correlation between acquisition premium and EPS from year t to t+2. All control variables are provided in the left column. Two premium measures (*Premium1* and *Premium2*) are used, while results are provided in columns (1)–(3) and (4)–(6), respectively. Only successful acquisitions are used. All variables are winsorized at the 1 and 99 percentiles. *T*-statistics are provided in parentheses. ***, **, and * denote significance at the 1%, 5% and 10% levels, respectively.

Control Variables	EPS_t	EPS_{t+1}	EPS_{t+2}	EPS_t	EPS_{t+1}	EPS_{t+2}
	(1)	(2)	(3)	(4)	(5)	(6)
Premium1	0.0200	−0.0087	−0.0966**			
	(0.4894)	(-0.1919)	(-2.3280)			
Premium2				−0.0043*	−0.0045**	−0.0070***
				(-1.8028)	(-2.0015)	(-2.9157)
CashPay	0.0554	0.0711	0.0248	0.0563	0.0446	0.0006
	(0.8598)	(1.1456)	(0.2904)	(0.9382)	(0.8604)	(0.0067)
Relevance	0.0477	0.0089	0.0410	0.0445	0.0420	0.0479
	(1.0691)	(0.1618)	(1.0683)	(1.1655)	(0.9765)	(0.7276)
MTB_Asset	0.0410***	0.0270**	−0.0145	0.0434	0.0203*	−0.0077
	(2.6144)	(2.7476)	(-0.7860)	(1.6873)	(1.9416)	(-0.6307)
MTB_Equity	−0.0054	−0.0046	0.0104*	−0.0030	0.0020	0.0139**
	(-0.7278)	(-1.0811)	(2.1813)	(-0.2571)	(0.7617)	(2.5061)
Growth	0.0650**	0.0409	0.0383	0.0635	0.0245	0.0327
	(2.5460)	(1.6437)	(0.9740)	(1.7046)	(1.1973)	(0.7718)
Leverage	−0.1751*	0.0070	−0.1194	−0.1632*	0.0789	−0.0555
	(-1.6873)	(0.1526)	(-0.9034)	(-1.9742)	(1.0548)	(-0.5226)
LnAsset	0.1605***	0.1275***	0.1484***	0.1604***	0.1270***	0.1616***
	(7.8204)	(5.4592)	(5.9159)	(10.0083)	(6.8108)	(3.3716)
Share	−0.0072	−0.0458	−0.0940	−0.0139	−0.0119	−0.1699
	(-0.0682)	(-0.3299)	(-1.4137)	(-0.3019)	(-0.1212)	(-0.8679)
ROA	1.7291***	1.1609***	0.9160	1.7454***	1.0979***	1.0102**
	(5.9164)	(4.6337)	(1.4538)	(4.0248)	(6.4860)	(2.5432)
OverConfidence	0.1130	0.0914	0.2038*	0.1116	0.1210	0.1715
	(1.2555)	(1.0359)	(2.1899)	(1.3636)	(1.5314)	(1.3444)
Cashflow	1.2926***	1.2174***	0.8412**	1.2874***	0.9090***	0.9220***
	(6.2804)	(4.1450)	(3.0042)	(6.0555)	(5.7491)	(3.8773)
SDAR	3.8734	3.8132	10.9022*	3.7332	5.8900**	13.4398**
	(1.5075)	(1.1715)	(2.1812)	(0.9016)	(2.4744)	(2.5346)
CAR	0.3816***	0.4726***	0.2000**	0.3972***	0.4450***	0.2725
	(4.3580)	(6.8314)	(2.3046)	(4.8607)	(4.8980)	(1.6831)
Intercept	−3.4031***	−3.2806***	−3.1199***	−3.4164***	−3.0654***	−3.4387***
	(-7.0277)	(-5.4477)	(-4.8856)	(-8.4315)	(-6.7149)	(-3.2330)
N	1207	1204	1030	1207	1204	1030
adj.R^2	0.330	0.224	0.179	0.331	0.233	0.171

overpay. As a result, those who overpay will underperform in the long run.

5. Conclusion

We study the impact of analyst coverage on acquisitions by using partial acquisition data in China. We use partial acquisitions in our study because we can directly examine the post-acquisition performance of the targets rather than of the merged firms (for targets' listings will remain post successful partial acquisition). In the short run, we find targets with higher analyst coverage to earn lower acquisition premium. Our results are robust to a battery of robustness checks. In the long run, targets with higher analyst coverage outperform targets with lower analyst coverage, while targets with higher acquisition premium underperform targets with lower acquisition premium, based on the future EPS post acquisition. Our results are consistent with our hypotheses.

Since firms in emerging markets like China tend to have higher growth, higher information asymmetry, and more serious stock overvaluation, analyst coverage can effectively reduce information asymmetry and speed up price discovery. In the short run, targets with higher analyst coverage earn lower acquisition premium in the form of price correction from the previous overvaluation. As a result, targets with high analyst coverage outperform targets with low analyst coverage post acquisition. Our findings indicate that analyst coverage can effectively speed up price discovery and promote market efficiency by reducing information asymmetry. In addition, our results indicate that such reduction of information asymmetry can help acquirers make better investment decisions and targets obtain external resources that are otherwise infeasible in the higher information asymmetry environment.

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Appendix AVariable Definition

Adjust	Adjust is the average of the analyst forecast revisions, and it is calculated as follows: Adjust = SUM [Analyst's consensus earnings forecast for each month (i.e., the average number of all analysts' latest earnings per share forecast for the target company for each month of the 360 days prior to the M&A date) - the consensus forecast for the previous month]/The closing price of the target company at the end of last month (Chan et al., 1996; Gleason & Lee, 2003; and Jegadeesh et al., 2004).
AnaDummy	During the 360-day period prior to the first M&A announcement, if any analyst publishes any earnings forecast for the target company, AnaDummy = 1. Otherwise AnaDummy = 0 (Chan & Hameed, 2006; Yu, 2008; McNichols & O'Brien, 1997; and James & Karceski, 2006).
AnaNum	The total number of analysts who issue any earnings forecast for the target company during the 360-day period prior to the first M&A announcement.
AnaTimes	The total number of earnings forecasts issued by all analysts for the target company during the 360-day period prior to the first M&A announcement.
CAR	The cumulative abnormal return [-42, −1] from the announcement date (Cheng et al., 2016).
Cashflow	Operating cash flow/total assets (Cheng et al., 2016).
CashPay	If the primary method of payment is cash, then CashPay = 1. Otherwise, CashPay = 0 (Jory et al., 2016).
EPS	Annual earnings per share.
Growth	($Revenue_t$ – $Revenue_{t-1}$)/$Revenue_{t-1}$ (Cheng et al., 2016).
Leverage	Total Liabilities/Total Assets (Li, Duan, He & Chan, 2018).
LnAsset	Natural log of the target firm's total assets is used to measure firm size (Beckman & Haunschild, 2002; Comment & Schwert, 1995; and Reuter et al., 2012).
MTB_Asset	According to Cheng et al. (2016), Market-to-Book Value of Asset = (Book Value of Liabilities + Total Market Value of Stocks)/Book Value of Total Assets.
MTB_Equity	Market-to-Book Value of Equity = Total Market Value of Stock/Shareholder's Equity (Cheng et al., 2016).
OverConfidence	According to Chen and Li (2016), Overconfidence = the Sum of the Compensations of the Top Three Highest Paid Executives/the Sum of the Compensations of All Senior Executives, Directors, and Supervisors.
Precision	In the first stage, $Precision_{i,t}$ = the average of the forecast accuracy of all analysts on company i within 360 days before the earnings announcement date. The forecast accuracy of each analyst = $(-1) \times \mid F_i - A_{i,t} \mid / P_{i,t-1}$, where Fi is the analyst's forecast of company i's earnings in the 360 days before the event date, Ai, t is the actual earnings per share of the company i in year t, and Pi, t-1 is the closing price of the stock the month before the analyst publishes the earnings forecast. In the second stage, Precision = 1 if the accuracy of the analyst's earnings forecast is greater than its mean from step one; otherwise, Precision = 0 (Duru & Reeb, 2002).
Premium1	Premium1 = (Offer Price Per Share – Closing Price Four Weeks Before the Announcement Date/the Closing Price Four Weeks Before the Announcement Date (Schwert, 1996; Reuter et al., 2012; and Jory et al., 2016). The four week time lag is used to ensure the baseline of the stock price is not affected by potential information leakage prior to the official announcement date (Nathan & O'Keefe, 1989).
Premium2	Premium2 = (Offer Price Per Share - Net Assets Per Share)/Net Assets Per Share. (Brau et al., 2010; Laamanen, 2007; Chen & Li, 2016; and Jory et al., 2016).
Proportion	Proportion of the target being acquired during the acquisition. It is obtained from the CSMAR.
Relevance	According to the CSMAR M&A database, if the M&A event is a related party transaction, Relevance = 1. Otherwise, Relevance = 0. Related-party transactions refer to transactions between related parties. The Ministry of Finance of China stipulates that if one party has the ability to directly or indirectly control the other in the financial and operating decisions of an enterprise, or exerts a significant influence on another party, they are considered as related parties. If two or more parties are under the control of one party, they will also be regarded as related parties (Li et al., 2018).
ROA	According to Cheng et al. (2016), ROA = Net Income/Total Assets.
SDAR	According to Cheng et al. (2016), SDAR is the standard deviation of the abnormal returns during [-365,-28] from the first announcement date, while daily abnormal return is calculated as the actual stock return of the target firm - market return.
Share	Shareholding ratio of the largest shareholder is downloaded from the China Stock Market and Accounting Research (CSMAR).
Transparency1	Following Hutton et al. (2009), we use earning management of the company to calculate the transparency index. Specifically, we first use the modified Jones model by industry (DeChow, Sloan, & Hutton, 1995) to calculate the nondiscretionary accruals for each firm by using a two-stage process. Stage 1: $\frac{TAC_{j,t}}{Asset_{j,t-1}} = \beta_0\left[\frac{1}{Asset_{j,t-1}}\right] + \beta_1\left[\frac{\Delta Sales_{j,t}}{Asset_{j,t-1}}\right] + \beta_2\left[\frac{PPE_{j,t}}{Asset_{j,t-1}}\right] + \varepsilon_{j,t}$ Stage 2: $DTAC_{j,t} = \frac{TAC_{j,t}}{Asset_{j,t-1}} - \widehat{\beta}_0\left[\frac{1}{Asset_{j,t-1}}\right] - \widehat{\beta}_1\left[\frac{\Delta Sales_{j,t} - \Delta Receivables_{j,t}}{Asset_{j,t-1}}\right] - \widehat{\beta}_2\left[\frac{PPE_{j,t}}{Asset_{j,t-1}}\right]$ Where TAC is the firm's total accruals, which is calculated as net income - cash flows from operations; ΔSale represents change in sales; ΔReceivables is change in accounts receivables; while PPE is property, plant and equipment. Next, we calculate transparency index as follows: Transparency = $(-1) \times$ [AbsV ($DTAC_{j,t-1}$) + AbsV ($DTAC_{j,t-2}$) + AbsV ($DTAC_{j,t-3}$)], where AbsV ($DTAC_{j,t-1}$) is the absolute value of firm j's nondiscretionary accruals in year t-1. Note that higher the transparency index indicates higher transparency.
Transparency2	Same as Transparency1. However, only information in year t-1 is used, rather than t-3 ~ t-1.

References

Allen, J., & Phillips, G. (2000). Corporate equity ownership, strategic alliances, and product market relationships. *The Journal of Finance, 55*, 2791–2815.

Allen, F., Qian, J., & Qian, M. (2005). Low, finance and economic growth in China. *Journal of Financial Economics, 77*, 57–116.

Andrade, G., Mitchell, M., & Stafford, E. (2001). New evidence and perspectives on mergers. *The Journal of Economic Perspectives, 15*, 103–120.

Asquith, P., Bruner, R. F., & Mullins, D. W. (1990). *Merger returns and the form of financing*. Working paper, Harvard University ((June)).

Barber, B., Lehavy, R., McNichols, M., & Trueman, B. (2006). Buys, holds, and sells: The distribution of investment banks' stock ratings and the implications for the profitability of analysts' recommendations. *Journal of Accounting and Economics, 41*, 87–117.

Beatty, R., & Ritter, J. (1986). Investment banking, reputation and underpricing of initial public offerings. *Journal of Financial Economics, 15*, 213–232.

Beckman, C. M., & Haunschild, P. R. (2002). Network learning: The effects of partners' heterogeneity of experience on corporate acquisitions. *Administrative Science Quarterly, 47*, 92–124.

Bostan, I., & Spatareanu, M. (2018). Financing innovation through minority acquisitions. *International Review of Economics & Finance*. https://doi.org/10.1016/j.iref.2018.02.007 (fourthcoming).
Bowen, R., Chen, X., & Cheng, Q. (2008). Analyst coverage and the cost of raising equity capital: Evidence from underpricing of seasoned equity offerings. *Contemporary Accounting Research, 25*, 657–700.
Bradley, J., Desai, A., & Kim, E. H. (1988). Synergistic gains from corporate acquisitions and their division between the stockholders of target and acquiring firms. *Journal of Financial Economics, 21*, 3–40.
Brau, J. C., Sutton, N. K., & Hatch, N. W. (2010). Dual-track versus single-track sell-outs: An empirical analysis of competing harvest strategies. *Journal of Business Venture, 25*, 389–402.
Brown, B., & Perry, S. (1994). Removing the financial performance halo from fortune's "most admired" companies. *Academy of Management Journal, 37*, 1347–1359.
Busom, I. (2000). An empirical evaluation of the effects of R&S subsidies. *Economics of Innovation and New Technology, 9*, 111–148.
Chan, K., & Hameed, A. (2006). Stock price synchronicity and analyst coverage in emerging markets. *Journal of Financial Economics, 80*, 115–147.
Chan, L. K., Jegadeesh, N., & Lakonishok, J. (1996). Momentum strategy. *The Journal of Finance, 51*, 1681–1713.
Cheng, P., Li, L., & Tong, W. H. S. (2016). Target information asymmetry and acquisition price. *Journal of Business Finance & Accounting, 43*, 976–1016.
Chen, S., & Li, W. (2016). Research on anchoring effect in M&A premium decision. *Chinese Economic Research, 6*, 114–127.
Chen, S. H., & Lu, C. C. (2017). Inter-business executive bonding on M&A premium. *Chinese Management World, 5*, 144–156.
Comment, R., & Schwert, G. (1995). Poison or placebo? Evidence on the deterrence and wealth effects of modern antitakeover measures. *Journal of Financial Economics, 39*, 3–43.
Covitz, D. M., & Harrison, P. (2003). *Testing conflicts of interest at bond ratings agencies with market anticipation: Evidence that reputation incentives dominate*. Finance and Economics Discussion Series 2003-68, US.
Das, S., Guo, R., & Zhang, H. (2006). Analysts' selective coverage and subsequent performance of newly public firms. *The Journal of Finance, 61*, 1159–1185.
De Long, J. B., Shleifer, A., Summers, L., & Waldmann, R. (1990). Positive feedback investment strategies and destabilizing rational speculation. *The Journal of Finance, 45*, 375–395.
DeChow, P. M., Sloan, R. G., & Hutton, A. (1995). Detecting earnings management. *The Accounting Review, 70*, 193–225.
Demiroglu, C., & Ryngaert, M. (2010). The first analyst coverage of neglected stocks. *Financial Management, 39*, 555–584.
Desai, H., Liang, B., & Singh, A. (2000). Do all-stars shine? Evaluation of analyst recommendations. *Financial Analysts Journal, 56*, 20–29.
Dhiensiri, N., & Sayrak, A. (2004). *The value effects of analyst coverage initiations*. Working Paper, University of Pittsburgh.
Diamond, D., & Verrechia, R. (1991). Disclosure, liquidity, and the cost of capital. *The Journal of Finance, 46*, 1325–1360.
Duru, A., & Reeb, D. M. (2002). International diversification and analysts' forecast accuracy and bias. *The Accounting Review, 77*, 415–433.
Ederington, I. H., & Yawitz, J. B. (1985). *The bond rating process*. Institute of Banking and Financial Markets, Washington University.
Fee, C., Hadlock, C., & Thomas, S. (2006). Corporate equity ownership and the governance of product market relationships. *The Journal of Finance, 61*, 1217–1251.
Feng, J. X. (2007). The transparency of information disclosure and securities of listed companies in China: Analyst prediction. *Chinese Financial Research, 6*, 136–147.
Feng, X., Hu, N., & Johasson, A. C. (2016). Ownership, analyst coverage, and stock synchronicity in China. *International Review of Economics & Finance, 45*, 79–96.
Franks, J. R., & Harris, R. S. (1989). Shareholder wealth effects of corporate takeovers: The UK experience 1955-85. *Journal of Financial Economics, 23*, 225–249.
Fuller, K., Netter, J., & Stegemoller, M. (2002). What do returns to acquiring firms tell us? Evidence from firms that make many acquisitions. *The Journal of Finance, 57*, 1763–1793.
Gleason, C., & Lee, C. (2003). Analyst forecast revisions and market price discovery. *The Accounting Review, 78*, 193–225.
Gregory, A. (1997). An examination of the long run performance of UK Acquiring Firms. *Journal of Business Finance & Accounting, 24*, 971–1002.
Gulati, R., & Higgins, M. C. (2003). Which ties matter when? The contingent effects of interorganizational partnerships on IPO success. *Strategic Management Journal, 24*, 127–144.
Hardford, J. (1999). Corporate cash reserves and acquisitions. *The Journal of Finance, 54*, 1969–1997.
Hayward, M. L. A., & Hambrick, D. C. (1997). Explaining the premiums paid for large acquisitions: Evidence of CEO hubris. *Administrative Science Quarterly, 42*, 103–127.
Healy, P. M., Hutton, A. P., & Palepu, K. G. (1999). Stock performance and intermediation changes surrounding sustained increases in disclosure. *Contemporary Accounting Research, 16*, 485–520.
Heckman, J. J. (1979). Sample selection bias as a specific error. *Econometrica, 47*, 153–161.
Hertzel, M., & Smith, R. (1993). Market discounts and shareholder gains for placing equity privately. *The Journal of Finance, 48*, 459–485.
Houston, J., & Ryngaert, M. (1997). Equity issuance and adverse selection: A direct test using conditional stock offers. *The Journal of Finance, 52*, 197–219.
Hutton, A., Marcus, A., & Tehranian, H. (2009). Opaque financial reports, R^2, and crash risk. *Journal of Financial Economics, 94*, 67–87.
Irvine, P. (2003). Incremental impact of analyst initiation of coverage. *Journal of Corporate Finance, 9*, 431–451.
Jackson, A. R. (2005). Trade generation, reputation, and sell-side analysts. *The Journal of Finance, 55*, 673–717.
James, C., & Karceski, J. (2006). Strength of analyst coverage following IPOs. *Journal of Financial Economics, 82*, 1–34.
Jegadeesh, N., Kim, J., Krische, S. D., & Lee, C. (2004). Analyzing the analysts: When do recommendations add value? *The Journal of Finance, 59*, 1083–1124.
Jory, S. M., Ngo, T. N., & Wang, D. (2016). Credit ratings and the premiums paid in mergers and acquisitions. *Journal of Empirical Finance, 39*, 93–104.
Jung, S. H. (2017). The contingent effect of analyst coverage: How does analyst coverage affect innovation and Tobin's Q? *Asian-Pacific Journal of Accounting & Economics, 24*, 43–67.
Kohers, N., & Kohers, T. (2001). Takeovers of technology firms: Expectations vs. Reality. *Financial Management, 30*, 35–54.
Krisgen, D. (2006). Credit ratings and capital structure. *The Journal of Finance, 61*, 1035–1072.
Krisgen, D. (2009). Do firms target credit ratings or leverage levels? *Journal of Financial and Quantitative Analysis, 44*, 1323–1344.
Laamanen, T. (2007). On the role of acquisition premium in acquisition research. *Strategic Management Journal, 28*, 1359–1369.
Lang, M., Lins, K., & Miller, D. (2004). Concentrated control, analyst following and valuation: Do analysts matter most when investors are protected least? *Journal of Accounting Research, 42*, 589–622.
Lang, M., & Lundholm, R. (1993). Cross-sectional determinants of analyst ratings of corporate disclosures. *Journal of Accounting Research, 31*, 246–271.
La Porta, R., Silanes, F. L., Shleifer, A., & Vishny, R. (1998). Law and finance. *Journal of Political Economy, 106*, 1113–1155.
Li, L., Duan, Y., He, Y., & Chan, K. C. (2018). Linguistic distance and mergers and acquisitions: Evidence from China. *Pacific-Basin Finance Journal, 49*, 81–102.
Liao, R. (2013). *What drives corporate block acquisitions? The case for financial constraints*. Working Paper, Rutgers Business School.
Lin, H. C., Chou, T. K., & Cheng, J. C. (2011). Does market misvaluation drive post-acquisition underperformance in stock deals? *International Review of Economics & Finance, 20*, 690–706.
Loderer, C., & Martin, K. (1992). Post-acquisition performance of acquiring firms. *Financial Management, 21*, 69–79.
Loughran, T., & Vijh, A. M. (1997). Do long-term shareholders benefit from corporate acquisitions? *The Journal of Finance, 54*, 165–202.
McNichols, M., & O'Brien, P. C. (1997). Self-selection and analyst coverage. *Journal of Accounting Research, 35*, 167–199.
McNichols, M., O'Brien, P. C., & Pamukcu, O. (2007). *Unaffiliated analysts' recommendations for IPO firms*. Working Paper, Stanford University.
Michaely, R., & Womack, K. L. (1999). Conflict of interest and the credibility of underwriter analyst recommendations. *Review of Financial Studies, 12*, 653–686.
Moeller, S. B., Schlingemann, F. P., & Stulz, R. M. (2004). Firm size and the gains from acquisitions. *Journal of Financial Economics, 73*, 201–228.
Morck, R., Yeung, B., & Yu, W. (2000). The information content of stock markets: Why do emerging markets have synchronous stock price movements? *Journal of Financial Economics, 58*, 215–260.
Nathan, K., & O'Keefe, T. B. (1989). The rise in takeover premiums: An exploratory study. *Journal of Financial Economics, 23*, 101–119.
Niden, C. (1988). The role of taxes in corporate acquisitions. *Journal of Accounting Research, 36*, 279–298.
Officer, M., Poulsen, A. B., & Stegemoller, M. (2009). Information asymmetry and acquirer returns. *Review of Finance, 13*, 467–493.
Ouimet, P. P. (2013). The society for financial studies. *Review of Financial Studies, 26*, 1021–1047.

Piotroski, J. D., Wong, T. J., & Zhang, T. (2015). Political incentives to suppress negative information: Evidence from Chinese listed firms. *Journal of Accounting Research, 53*, 405–459.

Reuter, J. J., Tong, T. W., & Wu, C. W. (2012). A signaling theory of acquisition premiums: Evidence from IPO targets. *Academy of Management Journal, 55*, 667–683.

Roll, R. (1986). The hubris hypothesis of corporate takeovers. *Journal of Business, 59*, 197–216.

Roulstone, D. (2003). Analyst following and market liquidity. *Contemporary Accounting Research, 20*, 552–578.

Sadka, R., & Scherbina, A. (2007). Analyst disagreement, mispricing, and liquidity. *The Journal of Finance, 62*, 2367–2403.

Schutte, M., & Unlu, E. (2009). Do security analysts reduce noise? *Financial Analysts Journal, 65*, 40–54.

Schwert, G. W. (1996). Markup pricing in mergers and acquisitions. *Journal of Financial Economics, 41*, 153–192.

Singhvi, S. S., & Desai, H. B. (1971). An empirical analysis of the quality of corporate financial disclosure. *The Accounting Review, 46*, 129–138.

Stuart, T. E., Hoang, H., & Hybels, R. (1999). Interorganizational endorsements and the performance of entrepreneurial ventures. *Administrative Science Quarterly, 44*, 315–349.

Travlos, N. (1987). Corporate takeover bids, methods of payment, and bidding firms' stock returns. *The Journal of Finance, 42*, 943–963.

Varaiya, N. P., & Ferris, K. R. (1987). Overpaying in corporate takeovers: The winner's curse. *Financial Analysts Journal, 43*, 209–219.

Walking, R. A., & Edmister, R. D. (1985). Determinants of tender offer premiums. *Financial Analysts Journal, 41*, 27–37.

Xu, N., Chan, K. C., Jiang, X., & Yi, Z. (2013). Do star analysts know more firm-specific information? Evidence from China. *Journal of Banking & Finance, 37*, 89–102.

Yu, F. (2008). Analyst coverage and earnings management. *Journal of Financial Economics, 88*, 245–271.